Computational Studies of Mixed Metal-Halide Perovskites

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A thesis submitted for the Honour School of Chemistry

Part II 2025-2026

Acknowledgements

I would like to express my gratitude to Professor Volker Deringer for his guidance, valuable insights, and direction throughout the course of this project. His expertise and feedback were really helpful in shaping the scientific direction of this work.

I would also like to thank my immediate supervisor, Bodoo Batnaran, whose continuous guidance, encouragement, and support have been invaluable in bringing the present study to completion. Their assistance throughout both the computational and analytical aspects of the project is greatly appreciated. Also, a thanks to all the members of VLD group for their insights through their presentations throughout the year as well as through feedback on my presentations.

Additionally, ChatGPT Edu was used during the course of this research as a supplementary academic and coding assistant. Specifically, it was used for assistance with coding and interpretation of software documentation, for obtaining initial literature direction and brainstorming research ideas, for example, evaluating the potential impact of research directions like the phase transitions, assistance in setting up the experiment for the different runs, and for reviewing sections of the thesis for improving language as well as for overseeing any grammatical corrections.

Summary

This thesis investigates the structural and finite-temperature behaviour of mixed-halide inorganic perovskites of composition $CsPb(Cl_{1-x}Br_x)_3$ using machine-learned interatomic potentials (MLIPs). Mixed-halide perovskites are of significant interest for optoelectronic applications due to their tunable structural and electronic properties; however, modelling these systems is computationally challenging because of substitutional disorder, anharmonic lattice dynamics, and composition-dependent phase behaviour.

To address these challenges, pretrained foundational MLIPs based on the MACE framework were benchmarked and subsequently fine-tuned using density functional theory (DFT) reference data generated for mixed-halide configurations. Symmetry-inequivalent structures were generated using the ICET framework and representative configurations were selected through CUR decomposition using SOAP descriptors in order to reduce configurational redundancy while retaining local structural diversity.

Initial benchmarking of foundational models demonstrated that the MACE-MPA potential provided reasonable agreement with experimental lattice parameters and reproduced the Vegard-like volume trends of the end-member compounds $CsPbCl_3$ and $CsPbBr_3$. However, it failed to accurately capture the expected octahedral tilting distortions. Fine-tuning the MPA model led to a substantial improvement in the prediction of equilibrium structural properties across the mixed-halide compositional range. The fine-tuned models showed improved agreement with experimental lattice parameters while preserving the Vegard-like trends in volume and other structural parameters. Furthermore, they successfully reproduced the correct octahedral tilt

patterns observed experimentally.

The effect of training-set size was investigated using CUR-selected datasets containing between 20 and 200 structures. The results showed that relatively small representative datasets were sufficient to reproduce equilibrium volume trends, suggesting that foundational MLIPs contain transferable chemical information which can be efficiently adapted through fine-tuning consistent with the previous studies. Equation-of-state analysis further showed that the fine-tuned models accurately reproduced the overall energy–volume landscape, yielding smooth and physically consistent curves that preserved the expected equation-of-state behaviour across the sampled volume range.

Octahedral tilting behaviour and finite-temperature molecular dynamics simulations were also examined in order to evaluate whether the fine-tuned potentials correctly captured the structural distortions characteristic of halide perovskites. The models reproduced the experimentally observed $a^-a^-c^+$ tilt pattern for the orthorhombic phases. Furthermore, NPT molecular dynamics simulations performed at 100 K, 250 K, and 400 K reproduced the experimentally expected structural phases at each temperature, demonstrating that the fine-tuned potentials capture the temperature-dependent phase behaviour of the material.

Overall, this work demonstrates that fine-tuned foundational MLIPs provide a computationally efficient framework for modelling mixed-halide perovskites. While similar methodologies exist for fine-tuning the foundational models, the performance of the foundational models hasn't been tested on the specific mixed halide perovskite system. Therefore, the results establish an initial route to validating the distortions and for performing large-scale finite-temperature simulations of compositionally disordered halide perovskites, which can provide insight into the structural dynamics of the $CsPb(Cl_{1-x}Br_x)_3$ system.

Contents

List of Abbreviations

ACE . . . . . . Atomic Cluster Expansion.

CG . . . . Conjugate Gradient minimization.

CUR . . . . . . Column–Row matrix decomposition method used for structure selection.

DFT . . . . . . Density Functional Theory.

EOS . . . . . . Equation of State.

FIRE . . . . . Fast Inertial Relaxation Engine minimization algorithm.

GGA . . . . . . Generalized Gradient Approximation.

HNF . . . . . . Hermite Normal Form.

ICET . . . . . Integrated Cluster Expansion Toolkit.

LAMMPS . . Large-scale Atomic/Molecular Massively Parallel Simulator.

MACE . . . . Message Passing Atomic Cluster Expansion.

MD . . . . . . Molecular Dynamics.

MLIP . . . . . Machine-Learned Interatomic Potential.

MPA . . . . . . Materials Project pretrained MACE foundation model.

PBEsol . . . . Perdew–Burke–Ernzerhof revised for solids exchange–correlation functional.

RMSE . . . . . Root Mean Square Error.

SNF . . . . . . Smith Normal Form.

SOAP . . . . . Smooth Overlap of Atomic Positions descriptor.

SQS . . . . . . Special Quasirandom Structure.

SVD . . . . . . Singular Value Decomposition.

XC . . . . Exchange–Correlation.

CASTEP . . . Cambridge Serial Total Energy Package.

PDynA . . . . Perovskite Dynamics Analysis Package.

1 Introduction

Metal halide perovskites have emerged as one of the most important classes of materials for next-generation optoelectronic applications, including solar cells, light-emitting diodes, lasers, photodetectors, and radiation detectors [1, 2]. Their popularity arises from their tunable band gaps, strong optical absorption, defect tolerance, and relatively low-cost synthesis routes [2–5].

In particular, all-inorganic lead halide perovskites such as $CsPbX_3$ (X = Cl, Br, I) have attracted considerable attention because of their improved thermal stability compared to hybrid organic–inorganic perovskites [6]. These materials crystallize in the ABX_3 perovskite structure where A is a monovalent cation (Cs^+), B is a divalent cation (Pb^{2+}), and X is a halide anion. The structure consists of corner-sharing PbX_6 octahedra forming a three-dimensional framework, with Cs ions occupying the cavities between the octahedra.The structural behaviour of halide perovskites is strongly influenced by the soft and anharmonic nature of the Pb–X framework, which gives rise to pronounced dynamic lattice fluctuations and octahedral tilting. Unlike more rigid oxide perovskites, these materials exhibit substantial temperature-dependent disorder and relaxational octahedral rotations [7].

For $CsPbX_3$ compounds, these structural distortions are closely connected to phase stability. Upon cooling, the materials typically undergo a sequence of symmetry-lowering phase transitions from a high-temperature cubic phase to tetragonal and eventually orthorhombic phases [8, 9]. These transitions are accompanied by cooperative octahedral rotations that alter the electronic structure, lattice dynamics, dielectric response, and charge-carrier behaviour of the material [10, 11]. Consequently, understanding finite-temperature structural behaviour is essential for improving device stability and predicting the operational performance of perovskite-based optoelectronic systems [12, 13].

Mixed-halide perovskites provide an additional degree of tunability by allowing compositional engineering of the halide sublattice. Partial substitution of one halide species with another modifies the unit cell volume, band gap, optical absorption edge, and structural stability. Mixed I/Br systems have therefore been widely investigated for photovoltaic applications because their electronic properties can be continuously tuned across the visible spectrum (band gap ranging from 1.73 eV for $CsPbI_3$ to 2.36 eV for $CsPbBr_3$) [14]. However, these materials often exhibit light-induced or thermally driven halide segregation, leading to local compositional inhomogeneities and long-term device degradation [15, 16].

Compared to mixed I/Br systems, mixed Br/Cl perovskites such as $CsPb(Cl_xBr_{1-x})_3$ have received substantially less theoretical attention despite their importance for wide-band-gap and blue-emitting optoelectronic applications (band gap for $CsPbCl_3$ lies around 2.98 eV) [17, 18]. Many existing studies on mixed Br/Cl systems primarily focus on static structural characterization and density functional theory (DFT) calculations of equilibrium properties such as lattice constants, unit-cell volumes, and compositional Vegard-like trends [19–21]. While the average structural behaviour is well described in these studies, probing the local and dynamic structural features is limited by configurationally disordered nature of the mixed halides.

Accurately modelling mixed-halide perovskites presents several interconnected challenges. First, substitutional disorder on the halide sublattice generates an extremely large configurational space, requiring either very large supercells or extensive sampling of many inequivalent atomic arrangements. For example, a 40-atom cubic $CsPb(Cl_xBr_{1-x})_3$ supercell contains eight formula units and therefore 24 halide sites. At $x = 0.5$, 12 of these sites are occupied by Cl and 12 by Br, giving

$$\binom{24}{12} = 2{,}704{,}156$$

possible Cl/Br arrangements before accounting for symmetry. The true configurational space is smaller because many arrangements are related by translational, rotational, or other space-group symmetry operations and therefore represent equivalent atomic configurations. Second, the Pb–X framework in halide perovskites exhibits strong anharmonic lattice dynamics, meaning that the potential energy surface is relatively flat and multiple distorted structures can exist at similar energies. As a result, small structural distortions or thermal fluctuations can significantly modify the relative stability of different phases and alter the preferred octahedral tilt patterns. Third, finite-temperature phase transitions involve collective lattice fluctuations and long-timescale dynamical processes that are difficult to access directly using conventional first-principles calculations due to their computational cost.

Machine-learned interatomic potentials (MLIPs) provide a promising route to overcome these limitations by combining near-DFT accuracy with the computational efficiency required for large-scale atomistic simulations [22]. Recent developments in equivariant message-passing neural network architectures, like the MACE [23] framework, have enabled the construction of highly accurate interatomic potentials capable of stable molecular dynamics simulations across chemically diverse systems.

Unlike conventional empirical force fields, which rely on predefined analytical forms and material-specific parameterization, MLIPs learn the potential-energy surface directly from quantum-mechanical reference data. For example, classical Buckingham force fields have been successfully developed for mixed-halide $CsPb(Br_xI_{1-x})_3$ perovskites through extensive fitting to DFT calculations. In contrast, MLIPs can represent complex local atomic environments and many-body interactions without prescribing a specific functional form, offering improved transferability across compositions, phases, and thermodynamic conditions [24, 25].

An important recent development is the emergence of pretrained foundational MLIPs trained on large and chemically diverse first-principles datasets [22]. Such models encode transferable representations of chemical bonding and local atomic environments that can subsequently be adapted to chemically specific systems through fine-tuning. This approach reduces the amount of expensive DFT data required to obtain accurate and transferable potentials while retaining physically meaningful representations learned during large-scale pretraining.

Although $CsPb(Cl_xBr_{1-x})_3$ has been studied experimentally and with DFT [20, 21, 26], there remains a gap for this mixed Br/Cl system, particularly for finite-temperature molecular dynamics, octahedral tilting, and composition-dependent phase-transition behaviour, which can be solved with systematically validated machine-learned interatomic potentials. Furthermore, the relationship between configurational disorder, octahedral tilts, and thermal phase stability in mixed Br/Cl systems remains poorly understood.

This work aims to develop and validate fine-tuned machine-learned interatomic potentials for $CsPb(Clx Br1-x)_3$ based on the MACE foundation-model framework. Particular emphasis is placed on evaluating whether the fine-tuned models can accurately capture the composition-dependent structural evolution of the mixed-halide system, including Vegard-like volume trends, octahedral tilt distortions, and reproduce the phases observed experimentally at different temperatures.

1.1 Aims of the Work

The primary aim of this work was to develop and validate machine-learned interatomic potentials (MLIPs) capable of describing the structural and finite-temperature behaviour of mixed-halide inorganic perovskites of composition $CsPb(Cl_{1-x}Br_x)_3$.

The specific objectives of the present work are:

- To benchmark several pretrained foundational MLIPs for predicting the equilibrium structural properties of $CsPbCl_3$ and $CsPbBr_3$.
- To investigate whether the fine-tuned model is able to reproduce the Vegard-like trend for lattice parameters and unit cell volume as the halide composition changes.
- To investigate whether the fine-tuned models correctly reproduce experimentally observed octahedral tilting patterns and structural distortions.
- To perform finite-temperature molecular dynamics simulations in order to study phase stability and thermally driven structural evolution in $CsPb(Cl_{1-x}Br_x)_3$ systems.

2
Methods

Contents

2.1 Density Functional Theory Methodology

2.1.1 Density Functional Theory

Density functional theory (DFT) calculations were performed in order to obtain first-principles reference data for the mixed-halide perovskite systems investigated in this work. DFT provides a quantum mechanical description of interacting electron systems by expressing the total energy as a functional of the electron density $\rho(\mathbf{r})$ rather than the many-body wavefunction [27]. Within the Kohn–Sham formalism, the total energy functional may be written as:

$$E[\rho] = T_s[\rho] + \int V_{\text{ext}}(\mathbf{r})\rho(\mathbf{r})d\mathbf{r} + E_{\text{H}}[\rho] + E_{\text{xc}}[\rho] \tag{2.1}$$

where $T_s[\rho]$ is the kinetic energy of the non-interacting electron system, V_{ext} is the external ionic potential, E_{H} is the Hartree electrostatic energy, and E_{xc} is the exchange–correlation functional[28].

The Kohn–Sham equations are solved self-consistently:

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V_{\text{eff}}(\mathbf{r})\right]\psi_i(\mathbf{r}) = \epsilon_i\psi_i(\mathbf{r}) \tag{2.2}$$

where $\psi_i(\mathbf{r})$ and ϵ_i are the Kohn–Sham orbitals and eigenvalues, respectively. The effective potential is given by

$$V_{\text{eff}}(\mathbf{r}) = V_{\text{ext}}(\mathbf{r}) + V_{\text{H}}(\mathbf{r}) + V_{\text{xc}}(\mathbf{r}) \tag{2.3}$$

where V_{ext} is the external ionic potential, V_{H} is the Hartree potential, and V_{xc} is the exchange–correlation potential. Electronic wavefunctions were represented using a plane-wave basis set [29, 30]:

$$\psi_i(\mathbf{r}) = \sum_{\mathbf{G}} c_{i,\mathbf{G}} e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} \tag{2.4}$$

where $\mathbf{G}$ denotes reciprocal lattice vectors and the summation is truncated according to a kinetic energy cutoff:

$$\frac{\hbar^2}{2m}|\mathbf{k}+\mathbf{G}|^2 \leq E_{\text{cut}} \tag{2.5}$$

Plane-wave approaches are particularly suitable for periodic crystalline solids such as halide perovskites due to their systematic convergence behaviour and compatibility with reciprocal-space methods .

Single-point DFT calculations were performed on the generated mixed-halide and the resulting total energies, atomic forces, and stress tensors were used as first-principles reference labels for machine-learning model fine-tuning and validation.

2.1.2 CASTEP Computational Setup

All density functional theory (DFT) calculations were performed using the CASTEP [31] plane-wave DFT package. CASTEP solves the Kohn–Sham equations for periodic systems using a plane-wave basis set together with pseudopotentials to describe the interaction between valence electrons and ionic cores. The use of periodic boundary conditions and Bloch wavefunctions makes the method particularly suitable for modelling crystalline solids.

Exchange–correlation effects were described using the Perdew–Burke–Ernzerhof revised for solids (PBEsol) generalized gradient approximation (GGA) functional [32]. PBEsol was selected because it provides improved predictions of equilibrium structural properties, including lattice parameters and volumes, for crystalline materials compared with standard PBE-GGA.

Owing to the periodic nature of crystalline solids, the electronic states are characterised by wavevectors $\mathbf{k}$ within the first Brillouin zone. Quantities such as the total electronic energy therefore require integration over reciprocal space, formally given by

$$\langle A \rangle = \frac{1}{\Omega_{\mathrm{BZ}}} \int_{\mathrm{BZ}} A(\mathbf{k})\, d\mathbf{k}, \tag{2.6}$$

where $A(\mathbf{k})$ represents a reciprocal-space quantity evaluated at wavevector $\mathbf{k}$ and Ω_{BZ} is the Brillouin-zone volume. In practice, CASTEP approximates this integral using a finite set of special k-points generated according to the Monkhorst–Pack scheme [33]. The accuracy of the calculation therefore depends on the density of the k-point sampling grid.

Convergence testing was performed with respect to both the plane-wave cutoff energy and the k-point sampling density. Parameters were considered converged when the difference in total energy between successive calculations was less than 1 meV atom^{-1}.

The converged DFT calculations provided the reference energies, and forces used for training and validation of the machine-learned interatomic potentials.

2.2 Configurational Sampling and Structure Generation

2.2.1 ICET Framework and Hart–Forcade Enumeration

Mixed-halide perovskites are substitutionally disordered systems in which Cl and Br species occupy the halide sublattice of the parent perovskite structure. Prior to symmetry reduction, the total number of distinguishable occupational configurations for a binary system is given by:

$$\Omega = \frac{N!}{N_\alpha! N_\beta!} \tag{2.7}$$

where N_α and N_β represent the number of atoms of species α and β, respectively, such that:

$$N = N_\alpha + N_\beta \tag{2.8}$$

Direct sampling of all configurations rapidly becomes computationally prohibitive due to the combinatorial growth of configuration space. Furthermore, many configurations are related by symmetry operations and are therefore symmetrically equivalent. Enumeration of symmetry-inequivalent configurations was performed using the ICET (Integrated Cluster Expansion Toolkit) framework [34], which implements the Hart–Forcade algorithm [35, 36].

Within the Hart–Forcade formalism, unique superlattices are generated from the parent lattice using Hermite normal form (HNF) transformation matrices, after which occupational configurations are assigned to lattice sites. Symmetry-equivalent superlattices are removed using the space-group symmetry operations of the parent crystal.

A superlattice is defined by an integer-valued Hermite normal form (HNF) matrix,

$$\mathbf{H} = \begin{pmatrix} h_{11} & 0 & 0 \\ h_{21} & h_{22} & 0 \\ h_{31} & h_{32} & h_{33} \end{pmatrix}, \tag{2.9}$$

where $h_{ii} > 0$ and $0 \leq h_{ij} < h_{jj}$ for $i > j$. The determinant of the HNF matrix, $\det(\mathbf{H})$, specifies the size of the supercell relative to the parent unit cell. Systematic enumeration of all symmetry-distinct HNF matrices therefore generates all unique superlattices of a given size. For example, an HNF matrix with $\det(\mathbf{H}) = 8$ generates a supercell containing eight times as many lattice sites as the parent unit cell.

Configurations related by a symmetry operation $g \in G$ are considered equivalent:

$$\sigma' = g(\sigma) \tag{2.10}$$

where σ and σ' denote atomic occupation configurations and G denotes the symmetry group of the parent lattice. Two configurations are equivalent if:

$$\exists\ g \in G \text{ such that } g(\sigma) = \sigma' \tag{2.11}$$

The Hart–Forcade algorithm additionally employs Smith normal form (SNF) matrices to identify translationally equivalent configurations efficiently through an algebraic treatment of translational symmetry. This approach enables systematic generation of all symmetry-inequivalent structures up to a chosen supercell size while avoiding redundant configurations.

In this work, the `enumerate_structures` functionality within ICET was used to generate mixed-halide configurations for selected compositions of the $CsPb(Cl_{1-x}Br_x)_3$ system. Disorder was restricted to the halide sublattice while the Cs and Pb sublattices remained fixed.

2.2.2 Special Quasirandom Structures (SQS)

In addition to exact enumeration approaches, substitutional disorder was also represented using the Special Quasirandom Structure (SQS) methodology [37]. The SQS approach aims to reproduce the ensemble-averaged correlation functions of an ideal random alloy within a finite periodic supercell.

For a binary alloy, the correlation function for a cluster α may be written as:

$$\Pi_\alpha = \frac{1}{N_\alpha} \sum_{i,j,\dots \in \alpha} \sigma_i \sigma_j \dots \tag{2.12}$$

where σ_i is an occupation variable associated with lattice site i, typically assigned values of $+1$ and -1 for the two atomic species in a binary alloy.

The SQS method constructs structures whose low-order correlation functions approximate those of a perfectly random alloy:

$$\Pi_\alpha^{\mathrm{SQS}} \approx \Pi_\alpha^{\mathrm{random}} \tag{2.13}$$

Unlike exact enumeration methods, SQS does not explicitly generate all symmetry-inequivalent configurations. Instead, it provides a statistically representative approximation to configurational disorder using finite-size periodic supercells. This makes the method particularly useful for modelling larger systems where exhaustive enumeration becomes computationally intractable. For example, in previous literature it has been demonstrated that an SQS containing only 16 atoms reproduced the band-gap properties of a 2304-atom random-alloy model of $Al_{0.5}Ga_{0.5}As$ to within 0.015 eV, highlighting the ability of SQS to capture the essential features of configurational disorder at a fraction of the computational cost [37].

In this work, $2 \times 2 \times 2$ supercell of the orthorhombic unit cell was constructed prior to relaxation. The use of a supercell serves several purposes. First, it reduces finite-size effects that may artificially constrain structural distortions in very small periodic cells. Second, larger supercells provide sufficient configurational freedom for collective octahedral tilting and local relaxation patterns to emerge naturally during optimisation. This is particularly important in halide perovskites, where structural distortions often involve correlated motions extending beyond a single primitive unit cell [38]. Finally, using a larger supercell improves the robustness of pressure and stress relaxation during geometry optimisation.

Structural relaxation was performed in LAMMPS [39] using anisotropic cell relaxation combined with the conjugate-gradient (CG) minimisation algorithm. Anisotropic relaxation was chosen so that all lattice vectors could vary independently during optimisation, allowing the simulation cell to fully respond to internal stresses and reproduce the equilibrium orthorhombic distortion. This is essential because constraining the cell shape could artificially suppress physically relevant strain or tilting modes. The conjugate-gradient method was employed because it is a robust and efficient approach for obtaining minimum-energy structures in atomistic simulations. Relaxations were done using different potentials until both the force norm approached zero and the residual pressure approached approximately 0 bar which were chosen as the convergence criteria for complete relaxation, ensuring that the structures corresponded to mechanically stable equilibrium configurations.

2.3 Dataset Construction and CUR Selection

2.3.1 CUR Selection

The large number of symmetry-inequivalent structures generated using ICET leads to substantial configurational redundancy, making direct DFT labelling computationally expensive. Although many supercells differ globally in their halide arrangements, they often contain very similar local atomic environments. Since machine-learned interatomic potentials (MLIPs) are fundamentally local models that learn atomic interactions from neighbouring coordination environments, repeatedly sampling structurally similar local configurations provides limited additional information for training while significantly increasing the computational cost of dataset generation. Consequently, constructing compact yet representative datasets requires selecting structures that maximize the diversity of local chemical environments and configurational motifs. To achieve this, CUR-based structure selection was employed prior to fine-tuning. [40].

Each mixed-halide configuration was first represented using Smooth Overlap of Atomic Positions (SOAP) descriptors [41], forming a feature matrix

$$X \in \mathbb{R}^{n\times d}, \tag{2.14}$$

where (n) denotes the number of structures and (d) denotes the dimensionality of the descriptor space. Each row of (X) therefore corresponds to a feature vector representing a particular atomic configuration. The SOAP descriptors were constructed by expanding the local atomic neighbour density in a basis of radial functions and spherical harmonics. Rotationally invariant power-spectrum coefficients were then computed and combined to produce a structure-level descriptor that captures both the local chemical composition and geometric arrangement of atoms. These descriptors provide a compact numerical representation of structural similarity and were used as the input features for the CUR-based structure selection procedure.

To identify representative structures, a truncated singular value decomposition (SVD) of the feature matrix was performed:

$$X \approx U_k \Sigma_k V_k^T \tag{2.15}$$

where $U_k \in \mathbb{R}^{n\times k}$ and $V_k \in \mathbb{R}^{d\times k}$ contain the leading left and right singular vectors, respectively, and $\Sigma_k \in \mathbb{R}^{k\times k}$ contains the largest singular values of the descriptor matrix.

Structure selection was performed using statistical leverage scores [40] derived from the truncated left singular-vector matrix U_k. The leverage score associated with structure i is given by

$$\ell_i = \|(U_k)_{i,:}\|_2^2 \tag{2.16}$$

where $(U_k)_{i,:}$ denotes the i-th row of U_k. Structures with large leverage scores correspond to configurations that contribute strongly to the dominant low-rank subspace of the dataset and therefore contain important structural information.

Representative structures were selected according to their leverage scores, thereby reducing configurational redundancy while retaining structurally diverse configurations across the compositional space.

The selected subset of configurations was subsequently used for DFT single-point calculations and machine-learning model fine-tuning. By reducing the number of redundant structures requiring first-principles labelling, the CUR selection procedure substantially lowers the computational cost associated with dataset generation.

2.4 Foundational Machine-Learned Interatomic Potentials and Fine-Tuning

2.4.1 MACE Foundational Models

Machine-learned interatomic potentials based on equivariant neural network architectures have recently demonstrated accuracy approaching that of first-principles calculations for predicting energies, forces, and stresses in crystalline materials [42]. In this work, the Message Passing Atomic Cluster Expansion (MACE) framework [23] was employed as the primary machine-learning architecture.

MACE combines equivariant message-passing neural networks with the Atomic Cluster Expansion (ACE) formalism [43]. In ACE, the atomic neighborhood density surrounding atom i is expanded as:

$$\rho_i(\mathbf{r}) = \sum_j \delta(\mathbf{r} - \mathbf{r}_{ij}) \tag{2.17}$$

where $\mathbf{r}_{ij}$ represents the relative position vector between atoms i and j, and the summation runs over neighboring atoms within a finite cutoff radius.

The local atomic environments are projected onto symmetry-adapted basis functions:

$$A_v = \langle \rho_i | \phi_v \rangle \tag{2.18}$$

where ϕ_ν denotes a set of symmetry-adapted radial and angular basis functions, allowing systematic construction of many-body features that can be incorporated into rotationally equivariant message-passing architectures.

A key feature of MACE is rotational equivariance, whereby vector and tensor quantities transform consistently under three-dimensional rotations while scalar quantities such as the total energy remain invariant. This allows the model to consistently describe directional atomic interactions and distorted local environments, which is important in halide perovskites due to their octahedral tilting and dynamic lattice distortions..

The primary motivation for employing foundational MLIPs in this work is their ability to provide transferable representations of interatomic interactions learned from large-scale DFT datasets. Training a machine-learning potential entirely from scratch would require substantial computational resources together with extensive high-quality DFT data for halide perovskites. Fine-tuning pretrained foundational models instead enables previously learned chemical knowledge to be adapted efficiently towards $CsPbX_3$ systems while reducing the amount of system-specific training data required. Foundational MACE models like MACE-MPA are trained on large first-principles materials datasets, including the Materials Project Trajectory (MPtrj) dataset and a subsampled version of the Alexandria dataset, comprising millions of DFT-calculated structures spanning a wide range of chemistries and crystal structures [44, 45]. These foundational models capture transferable chemical and structural relationships learned across diverse classes of inorganic materials.

In the present work, the pretrained `medium-mpa-0` (labelled as MPA in the text) foundational MACE model was employed and subsequently fine-tuned for the $CsPb(Cl_{1-x}Br_x)_3$ system using DFT-labelled structures generated within the present workflow.

2.4.2 Model Fine-Tuning Strategy and graph-pes

Fine-tuning of pretrained MACE models was performed using the `graph-pes` package [23, 46], which provides infrastructure for training and evaluating graph neural network based machine-learned interatomic potentials.

The fine-tuning procedure consisted of initializing the model using pretrained foundational weights and subsequently optimizing the parameters using DFT-labelled mixed-halide structures. Pretrained MACE foundation models were employed as the starting point for transfer learning.

Training was performed using both total energies and atomic forces as target quantities. The total loss function was constructed from weighted energy and force contributions:

$$\mathcal{L} = w_E \mathcal{L}_E + w_F \mathcal{L}_F \tag{2.19}$$

where the per-atom energy loss is given by:

$$L_E = \frac{1}{N_{\text{struc}}} \sum_i \left(\frac{E_i^{\text{pred}}}{n_i} - \frac{E_i^{\text{DFT}}}{n_i} \right)^2 \tag{2.20}$$

and the force loss is defined as:

$$\mathcal{L}_F = \frac{1}{3N_{\text{atoms}}} \sum_i \left| \mathbf{F}_i^{\text{pred}} - \mathbf{F}_i^{\text{DFT}} \right|^2 \tag{2.21}$$

where w_E and w_F denote the relative weighting factors for the energy and force contributions, respectively.

Fine-tuning was performed using the AdamW optimizer [47] with a learning rate of 10^{-5} together with a ReduceLROnPlateau learning-rate scheduler. A cutoff radius of 6 Å was employed for local atomic environments. Training and validation datasets were provided in `extxyz` format and were divided into separate training and validation subsets (80:20 ratio).

Early stopping based on the validation force RMSE was employed to avoid overfitting during training. The resulting fine-tuned models were subsequently validated against DFT calculations and experimental structural trends for mixed-halide perovskites.

3
Results

Contents

3.1 Foundational Models and Structural Benchmarking

3.1.1 Selection of Foundational Machine-Learning Interatomic Potentials

In this work, four pretrained machine-learned interatomic potentials (MLIPs) were selected for benchmarking and subsequent fine-tuning studies: MACE-MPA, MACE-MP-0a, MACE-OMAT-0 (referred as OMAT) [48], and MACE-MP-0b3 (referred as OB3) [23]. These models were chosen because they represent different approaches to universal atomistic modelling, with differences in neural-network architecture, training datasets, and pretraining strategies.

Benchmarking the zero-shot, *i.e.*, without any additional fine-tuning on mixed halide perovskite data, structural performance of these pretrained models provides an important first step in identifying physically reliable starting points for subsequent fine-tuning and mixed-composition simulations. The comparison additionally enables systematic assessment of deviations from experi- mentally observed structural trends.

3.1.2 End-Member Structural Properties

The performance of several foundational machine-learning interatomic potentials (MLIPs) is summarized in Table 3.1. The end-member compounds $CsPbCl_3$ and $CsPbBr_3$ were selected as reference systems as an initial test, enabling direct comparison between predicted and measured lattice parameters.

The orthorhombic phase was used as the starting structure for both compounds because it represents the experimentally stable low-temperature phase of $CsPbX_3$ (X = Cl, Br). Previous experimental and theoretical studies have shown that $CsPbCl_3$ and $CsPbBr_3$ adopt distorted orthorhombic perovskite structures at low temperature due to octahedral tilting of the PbX_6 framework, which lowers the symmetry relative to the ideal cubic perovskite structure [49, 50]

(a) $CsPbCl_3$

	Lattice parameters											
	a (Å)	Δ (%)	b (Å)	Δ (%)	c (Å)	Δ (%)	α (°)	Δ (%)	β (°)	Δ (%)	γ (°)	Δ (%)
Experimental	7.899 (14)	-	11.247 (11)	-	7.901 (14)	-	90.0	-	90.0	-	90.0	-
mpa	7.911	0.15	11.342	0.84	8.191	2.54	90.0	0.0	90.0	0.0	90.0	0.0
omat-0	7.951	0.65	11.395	1.31	8.132	2.91	90.0	0.0	90.0	0.0	90.1	0.1
mp-0a	7.956	1.22	11.388	1.32	8.142	2.82	90.0	0.0	90.0	0.0	90.0	0.0
ob3	7.937	0.98	11.398	1.41	8.132	2.69	90.0	0.0	90.0	0.0	90.0	0.0

(b) $CsPbBr_3$

	Lattice parameters											
	a (Å)	Δ (%)	b (Å)	Δ (%)	c (Å)	Δ (%)	α (°)	Δ (%)	β (°)	Δ (%)	γ (°)	Δ (%)
Experimental	8.203 (18)	-	11.753 (22)	-	8.250 (21)	-	90.0	-	90.0	-	90.0	-
mpa	8.273	0.85	11.896	1.21	8.481	2.80	90.0	0.0	90.0	0.0	90.0	0.0
omat-0	8.23	0.32	11.908	1.32	8.550	3.63	90.0	0.0	90.0	0.0	90.0	0.0
mp-0a	8.276	0.88	11.97	1.84	8.536	3.46	90.0	0.0	90.0	0.0	90.0	0.0
ob3	7.937	3.25	11.47	2.41	8.065	2.24	90.0	0.0	90.0	0.0	90.0	0.0

Table 3.1: Comparison of experimentally reported and MLIP-relaxed lattice parameters for orthorhombic $CsPbCl_3$ and $CsPbBr_3$ obtained using different foundational machine-learning interatomic potentials (MLIPs). Structural relaxations were performed on $2 \times 2 \times 2$ orthorhombic supercells using anisotropic cell relaxation and conjugate-gradient minimisation. The tables report the relaxed lattice constants (a, b, and c), lattice angles (α, β, and γ), and the corresponding percentage deviations (Δ) relative to experimental reference values.

The relaxed lattice parameters obtained from each MLIP were then compared against experimental reference values in Table 3.1. Lattice constants provide a physically meaningful benchmark because they are determined by the equilibrium separation between atoms predicted by the potential. Deviations from the experimental lattice parameters therefore indicate inaccuracies in the predicted equilibrium geometry and may reflect systematic overbinding or underbinding tendencies within the model.

Several important observations emerge from the results. First, all models successfully preserve the orthogonal lattice geometry, with α, β, and γ remaining close to 90°. In contrast, larger discrepancies are observed in the lattice constants a, b, and especially c. This suggests that the dominant source of error may arise from inaccuracies in the predicted equilibrium bonding geometry, such as bond distances, octahedral tilting, or cell strain.

Across both $CsPbCl_3$ and $CsPbBr_3$, the MACE-MPA model consistently yields the lowest overall deviations from experiment. In particular, its lattice parameter errors remain comparatively small and balanced across all crystallographic directions.

By contrast, some models perform reasonably well for one composition but show larger deviations for the other. This reduced consistency suggests that the corresponding potentials do not transfer equally well between chemically similar compositions. Since transferability is critical for subsequent mixed-composition modelling and fine-tuning, models that exhibit stable performance across multiple end-member compositions are preferable starting points.

An additional trend observed in both systems is the systematic expansion of the lattice parameters relative to experiment, particularly along the *c*-axis. This may indicate a slight underestimation of attractive interactions, or an overestimation of short-range interatomic repulsion between the atoms, resulting in larger equilibrium lattice parameter.

The larger errors observed for $CsPbBr_3$ compared to $CsPbCl_3$ in several models may also reflect the increased structural softness and polarisability associated with the larger bromide ion. Bromide-containing perovskites generally exhibit more anharmonic lattice dynamics [51] and weaker bonding stiffness than chloride analogues, which may contribute to the larger structural deviations observed in several transferable interatomic potentials.

Overall, the benchmarking results demonstrate that while all tested MLIPs reproduce the qualitative orthorhombic framework *ie.* angles close to 90°, significant differences exist in quantitative accuracy.

3.1.3 Volume–Composition Relationships

An important structural characteristic of mixed-halide perovskites is the variation of equilibrium lattice volume with halide composition. To further evaluate the transferability of the selected foundational MLIPs beyond the end-member compounds, the relaxed cell volumes of $CsPbCl_{3-x}Br_x$ were analysed as a function of bromide concentration and compared against both experimental measurements and DFT reference data, as shown in Figure 3.1. The approximately linear increase in equilibrium volume with bromide concentration demonstrates Vegard-like behaviour across the mixed-halide series. Experimental measurements for samples obtained via mechanosynthesis are compared with DFT calculations, which show similar compositional trends but systematic quantitative differences in equilibrium volume. The experimentally reported trend is reproduced from previous literature studies investigating Vegard-like behaviour in mixed-halide cesium lead perovskites [21]

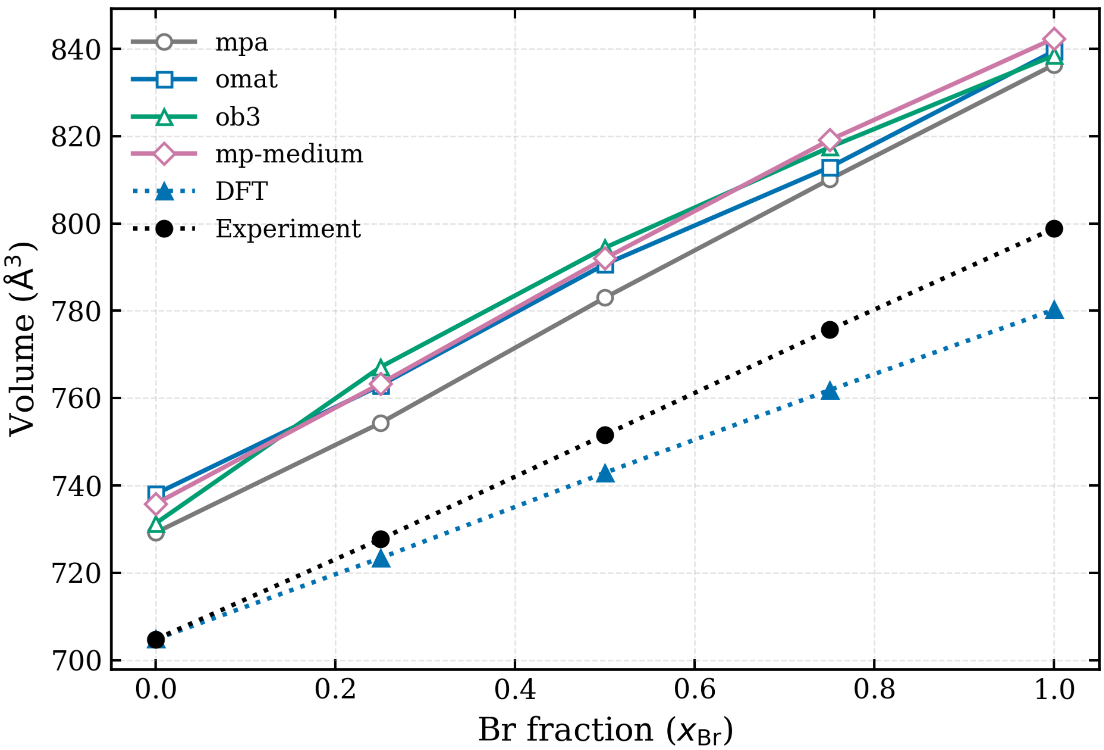

Figure 3.1: Variation of relaxed cell volume with bromide composition for $CsPbCl_{3-x}Br_x$ predicted using different foundational machine-learning interatomic potentials (MLIPs), together with DFT and experimental reference data. Note: For brevity, the models MACE-MPA, MACE-MP-0a, MACE-MP-0b3, and OMAT are hereafter referred to as mpa, mp-medium, ob3, and omat in the figure captions, respectively.

Vegard's law describes the approximately linear variation of lattice parameters or unit-cell volume with composition in solid solutions. For mixed-halide perovskites, increasing bromide concentration is expected to increase the equilibrium cell volume

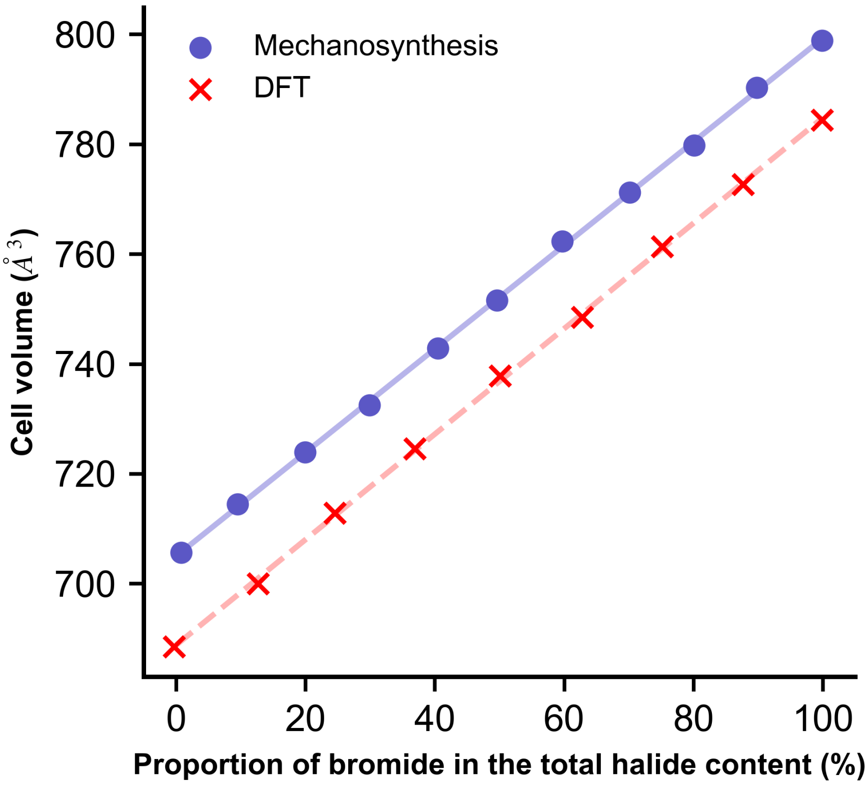

Figure 3.2: Experimentally measured and DFT-predicted variation of the unit-cell volume with bromide composition in $CsPbCl_{3-x}Br_x$. The figure compares cell volumes obtained from mechanosynthesised samples and DFT calculations across the mixed-halide compositional range. Reproduced from Ref. [21].

because Br^- possesses a larger ionic radius than Cl^-. Consequently, substitution of Cl by Br expands the Pb–X bond lengths and leads to systematic lattice expansion across the alloy series. Previous experimental and computational studies on $CsPbCl_{3-x}Br_x$ systems have reported near-linear volume expansion with increasing bromide fraction, consistent with Vegard-like behaviour [21].

The experimental data shown in Figure 3.2 exhibit an approximately linear increase in cell volume from $CsPbCl_3$ to $CsPbBr_3$, indicating that the mixed-halide alloys maintain continuous solid-solution behaviour across the investigated composition range. The DFT reference calculations also reproduce a similar linear trend, although the predicted volumes remain systematically lower than experiment throughout the composition range. Such deviations are commonly observed in DFT calculations and may arise from the choice of exchange–correlation functional, which can influence equilibrium bond lengths and lattice volumes.

All foundational MLIPs considered in this work successfully reproduce the qualitative Vegard-like linear increase in cell volume with bromide concentration. This is an important result because it demonstrates that the models correctly capture the overall compositional dependence of structural expansion within the alloy system. In particular, the approximately linear behaviour suggests that the models preserve the underlying chemical trend associated with halide substitution rather than introducing artificial structural discontinuities.

However, significant quantitative differences are observed between the MLIP predictions and the experimental reference data. All tested MLIPs systematically overestimate the equilibrium cell volumes across the entire composition range. This behaviour is consistent with the end-member structural benchmarking presented previously, where the relaxed lattice constants obtained from the MLIPs were generally larger than the experimental values for both $CsPbCl_3$ and $CsPbBr_3$.

Among the evaluated models, MPA exhibits the closest agreement with experiment across the full alloy series. Although the predicted volumes remain larger than experiment, the deviation is consistently smaller than for OMAT, OB3, and MACE-MP-0a. This observation is again consistent with the earlier end-member analysis, where MPA produced the lowest lattice parameter errors for both $CsPbCl_3$ and $CsPbBr_3$. The improved performance of MPA therefore appears to extend beyond the pure compounds and remains relatively consistent throughout intermediate alloy compositions.

The results therefore suggest that while further refinement is necessary to improve quantitative accuracy, the foundational models already contain transferable chemical information relevant to mixed-halide structural behaviour.

Overall, the volume–composition analysis reinforces the conclusions obtained from the end-member structural benchmarking. MPA demonstrates the best balance between quantitative agreement and consistent performance across different compositions, making it the most promising starting point for subsequent fine-tuning towards $CsPbCl_{3-x}Br_x$ alloy systems.

3.2 Fine-Tuning and Equilibrium Structural Properties

3.2.1 Fine-Tuning Strategy

Following the zero-shot benchmarking of the foundational models, fine-tuning was carried out using MACE-MPA as the starting point.

The purpose of fine-tuning was to reduce errors for the end-member structures and the mixed-composition $CsPb(Cl_{1-x}Br_x)_3$ system, the main subject of our study. This is important because mixed-halide perovskites contain many possible local Cl/Br arrangements, and a model that performs well only for $CsPbCl_3$ or $CsPbBr_3$ may not necessarily describe intermediate compositions reliably. Fine-tuning therefore provides a way of adapting the general chemical information contained in the pretrained model towards the specific structural environments present in the mixed-halide alloy.

The training structures were selected from the enumerated configuration space using CUR decomposition. This selection strategy was used because many mixed-halide configurations are structurally similar or symmetry-related, and training on redundant structures would increase computational cost without necessarily improving the information content of the dataset. In this work, fine-tuned models were trained using subsets of 20, 50, and 200 structures in order to test how sensitive the resulting equilibrium structural properties are to training set size.

The fine-tuned models were evaluated primarily through their predicted volume–composition relationships and relaxed lattice parameters.

3.2.2 Results of Fine-Tuning and Varying Training Set

The first question that was addressed was how the volume–composition relationship changes relative to the zero-shot foundational model. This comparison is important because the pretrained models already reproduced the qualitative increase in volume with Br fraction, but they systematically overestimated the absolute equilibrium volumes. Fine-tuning therefore tests whether system-specific DFT data can correct this volume offset while preserving the expected Vegard-like compositional trend.

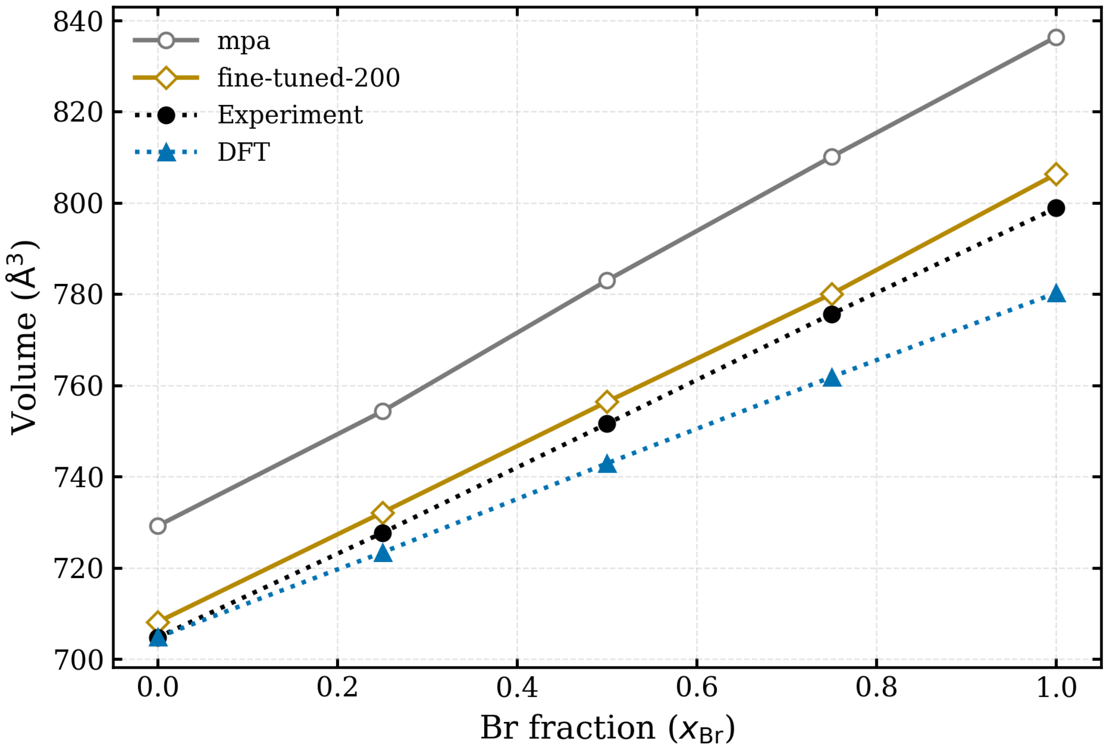

Figure 3.3: Comparison of zero-shot MPA and 200-shot fine-tuned MPA predictions for the equilibrium cell volume as a function of bromide fraction, x_{Br}. Note the improvement achieved using fine-tuning and the large deviations that occurs at higher Br concentration.

Figure 3.3 compares the volume–composition trends obtained from fine-tuned models trained on 200 structures. The fine-tuned model reproduces the near-linear increase in volume with increasing Br fraction, consistent with Vegard-like behaviour. More importantly, the fine-tuned curve is shifted substantially closer to the DFT and experimental references compared with the zero-shot foundational models.

This shows that fine-tuning corrects a large part of the equilibrium-volume bias potentially present in the pretrained potential.

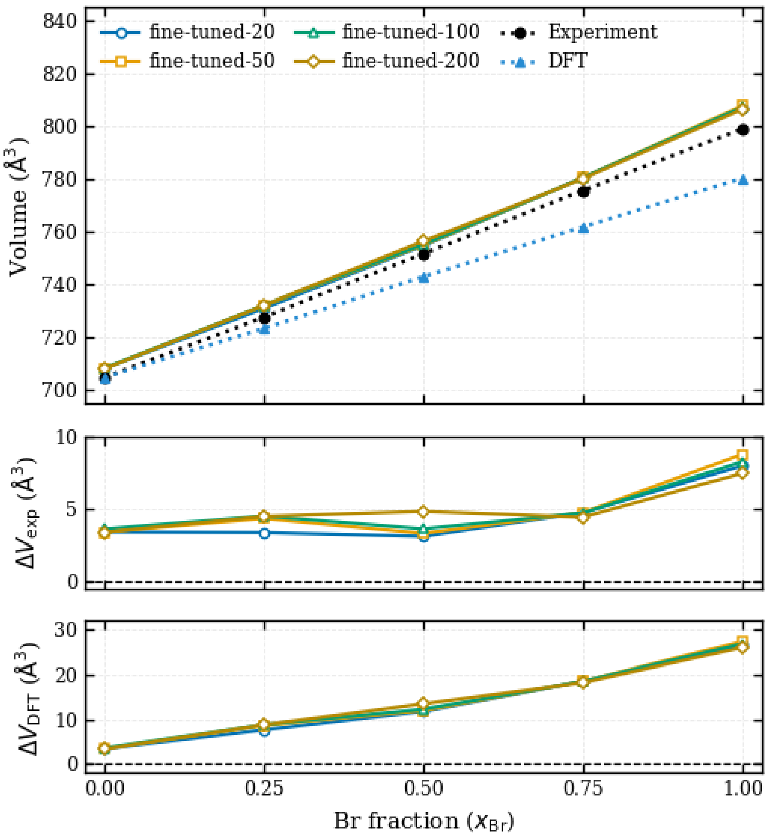

Figure 3.4: Equilibrium cell-volume predictions from fine-tuned MPA models as a function of bromide fraction, x_{Br}. The upper panel compares the absolute volumes from models trained with 20, 50, 100, and 200 fine-tuning examples against experimental and DFT reference values. The middle and lower panels show the corresponding residuals relative to experiment, a baseline is drawn for easier comparison. Here, $\Delta V_{\mathrm{exp}} = V_{\mathrm{model}} - V_{\mathrm{exp}}$, and DFT, $\Delta V_{\mathrm{DFT}} = V_{\mathrm{model}} - V_{\mathrm{DFT}}$, respectively.

After establishing the effect of fine-tuning itself, the second question was whether changing the number of training structures significantly changes the predicted equilibrium volumes. From Figure 3.4, the 20-, 50-, 100-, and 200-shot models give very similar curves across the full composition range. This indicates that the dominant information required to reproduce the volume trend is already captured by the smaller CUR-selected datasets. However, there is a discrepancy in regards to the greater deviation from the DFT data as compared to the experimental data, particularly for high Br composition. This has been discussed in section 2.2.5.

The small-shot results provide a further indication of compositional transferability. The models trained on reduced datasets, including very small training sets, still reproduce the overall volume trend across mixed compositions. This is notable because the smaller datasets do not explicitly contain all possible local Cl/Br arrangements. A possible interpretation is that the MPA foundation model was pretrained on the MPtrj dataset derived from the Materials Project database, it likely encountered compounds closely related to the Cs–Pb–Cl–Br perovskite family, including lead-halide and alkali-halide chemistries. As a result, the pretrained model is expected to have a useful representation of the bonding environments relevant to $CsPb(Cl_{1-x}Br_x)_3$, while the fine-tuning data adjusts this representation towards the mixed-halide system in the study.

The similarity between the 20-, 50-, 100-, and 200-shot models therefore suggests diminishing returns with increasing training-set size for this specific property. This should not be interpreted as evidence that larger datasets are unnecessary in general. Equilibrium volume is a bulk structural property, whereas forces, stresses, octahedral tilting, phonons, defect energetics, and finite-temperature phase behaviour are more sensitive to local atomic environments. These properties may require broader configurational sampling than is needed to reproduce the volume trend alone.

Overall, the results show that fine-tuning is effective in correcting the systematic volume overestimation of the foundational model, while increasing the training set beyond a relatively small representative subset has only a minor effect on equilibrium volumes. The close agreement between the different fine-tuned models supports the use of compact CUR-selected datasets for equilibrium structural fine-tuning, while also motivating further validation for properties that depend more strongly on local structure and dynamics.

As an additional check whether the Vegard-like trend is observed only for the volume or individual cell parameters as well, as a volume trend alone could, in

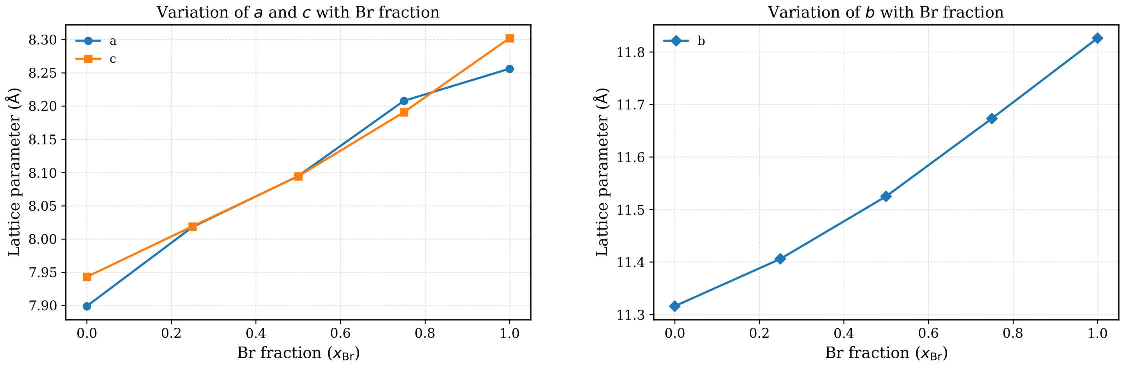

Figure 3.5: Variation of lattice parameters (a, b, and c in Å) as a function of bromide composition, x_{Br}, in mixed-halide perovskites. The left panel shows the evolution of lattice constants a and c, while the right panel shows the variation of b. Note the approximately linear composition dependence across the three lattice parameters.

principle, hide compensating errors between different crystallographic directions.

The lattice-parameter analysis provides an additional check beyond total volume. Figure 3.5 shows that the individual lattice parameters also increase with Br content. This supports the conclusion that the model predicts physically smooth structural expansion rather than producing an artificial volume trend from anisotropic distortions. The approximately linear behaviour of a, b, and c therefore strengthens the interpretation that the fine-tuned potential captures the equilibrium structural response to halide substitution.

3.2.3 Relaxation Protocols and Residual Pressure

A separate analysis of relaxation protocols was necessary because equilibrium lattice parameters and volumes are directly sensitive to the final stress state of the relaxed structure. During the relaxations, a discrepancy was observed: not all cell relaxations using cg style minimizations led to near zero pressure. In principle, a non-zero residual pressure after relaxation can shift the predicted equilibrium volume, even if the atomic forces are small. Therefore, before interpreting differences between DFT, experiment, and MLIP predictions, it was necessary to check whether the observed volume trends were genuine model predictions or artefacts of incomplete cell relaxation.

The initial relaxation protocol used energy minimisation with cell relaxation, but small residual pressures remained in some relaxed structures. To test whether these residual pressures affected the conclusions, alternative protocols were explored, including a relatively slow quenching protocol (5K/ps) and combined minimisation approaches involving CG and FIRE (as in Fig 3.6). The purpose of these additional calculations was not to repeatedly rerun the minimisation in pursuit of exactly zero pressure for structures that did not fully converge, but to determine whether reasonable changes in the relaxation pathway produced meaningful changes in the final volume–composition relationship.

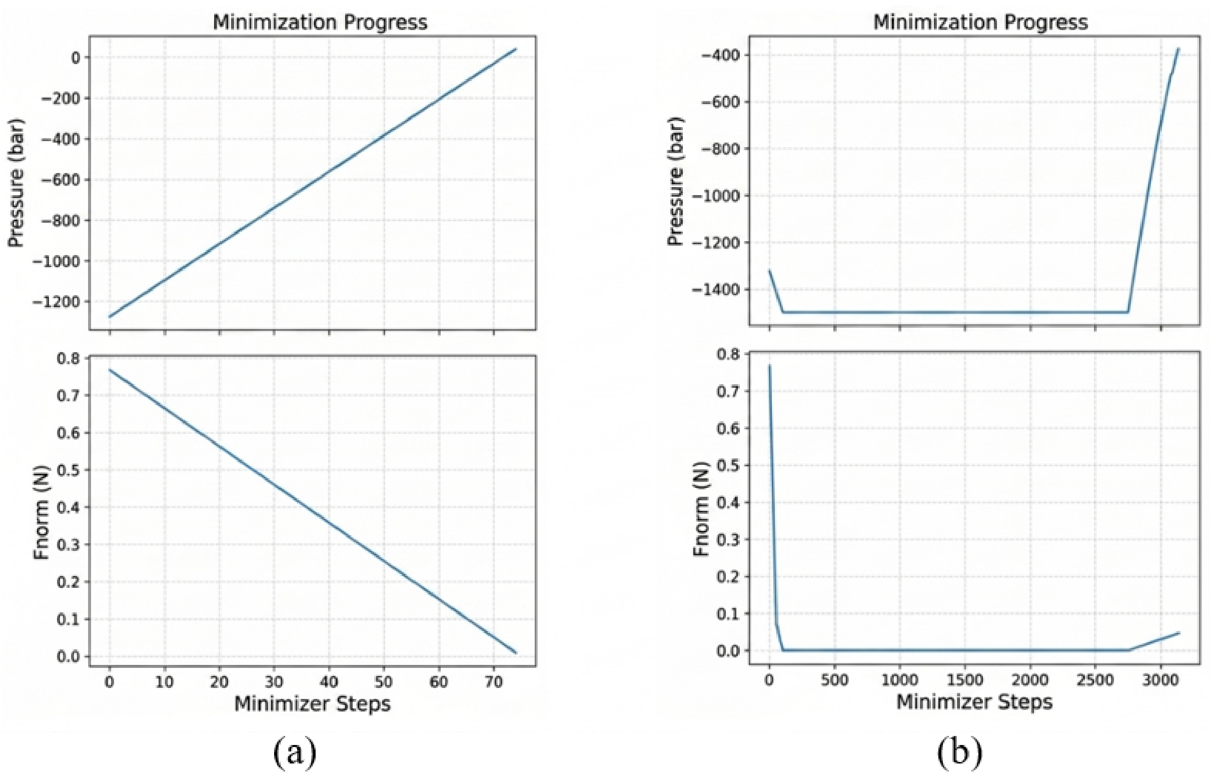

Figure 3.6: Comparison of energy minimization strategies for structural relaxation. (a) Conjugate gradient (CG) minimization, showing a gradual reduction in forces and corresponding change in pressure over simulation timesteps. (b) FIRE minimization applied first to relax atomic positions, followed by box relaxation. The plots illustrate behavior in terms of pressure and maximum force during the relaxation process for both minimizations.

The comparison between relaxation protocols in Fig 3.7 shows that slow quenching and slow quenching followed by CG/FIRE minimisation produce volume–composition

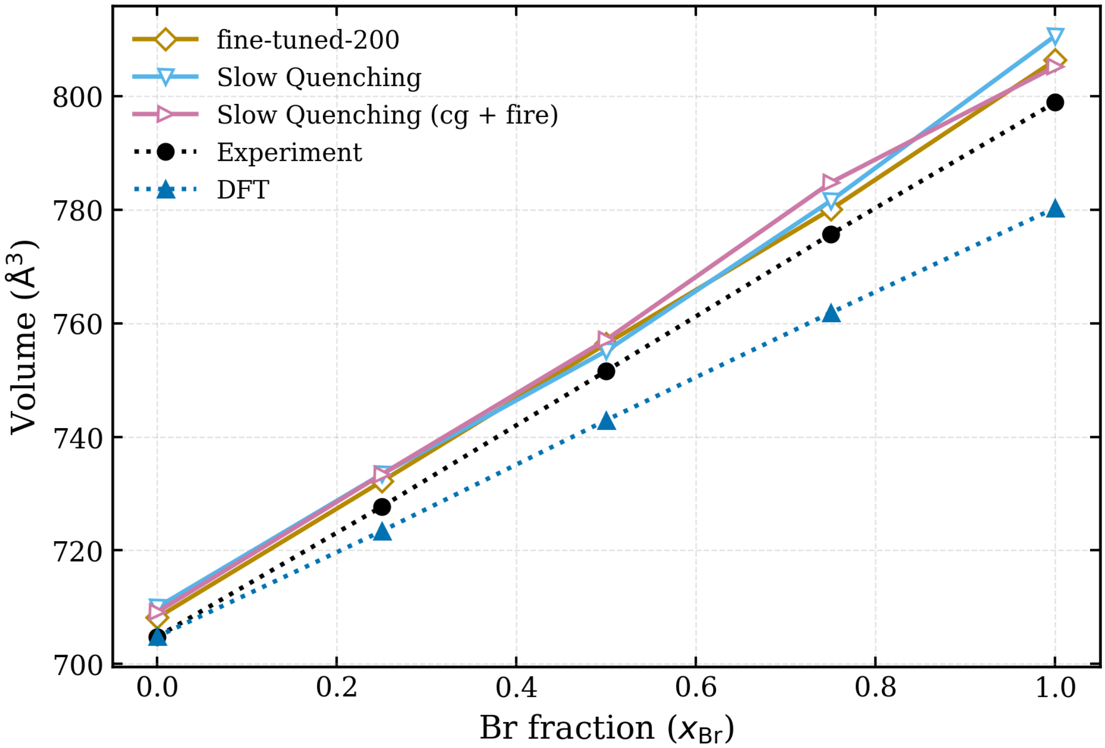

Figure 3.7: Unit cell volume (Å³) as a function of bromide composition (xBr) for structures obtained using different relaxation protocols: slow quenching and slow quenching combined with CG+FIRE minimization. Results are compared with experimental data and DFT calculations.

trends that are very similar to the original fine-tuned model results. This indicates that the small residual pressures remaining after relaxation do not significantly alter the equilibrium volume trends. In other words, the main conclusions are robust with respect to the relaxation protocol. Without this check, any offset between the fine-tuned model, DFT, and experiment could be attributed to incomplete relaxation rather than to the learned potential energy surface. By showing that different relaxation strategies lead to nearly the same volume trend, the analysis supports the conclusion that the predicted structural behaviour is a property of the fine-tuned model itself, rather than a numerical artefact of a particular minimisation route.

3.2.4 Equation-of-State Analysis

An equation of state (EOS) analysis was performed to investigate the volume differences between DFT and the fine-tuned model. Structures with inclrementally scaled volumes were generated and relaxed using the MLIP. The resulting structures were then evaluated with DFT to construct and compare the energy–volume curves. This analysis was carried out for $CsPbBr_3$, as it exhibited the largest discrepancy between the DFT and MLIP equilibrium volumes in the earlier structural validation results.

Volume–composition plots show only the final relaxed structures; they do not reveal whether the model has learned a physically reasonable energy landscape around equilibrium. Two models could predict different equilibrium volumes for very different reasons: one might have a physically consistent energy minimum shifted relative to DFT, while another might misrepresent the underlying energy–volume relationship through an incorrect curvature (bulk modulus) or unphysical behaviour under compression and expansion, such as discontinuities or sharp features in the EOS curve. The above analysis distinguishes between these possibilities. For $CsPbBr_3$, both the DFT and fine-tuned model EOS curves exhibit well-defined minima as in Fig 3.8, indicating that each method predicts a stable equilibrium volume. The fine-tuned model minimum occurs at a larger volume than the DFT minimum, which is consistent with the earlier volume–composition results. This confirms that the larger fine-tuned volume is not simply caused by a failed relaxation, but instead corresponds to the minimum of the fine-tuned model's potential energy surface.

The curvature of the EOS curve near the minimum is related to the bulk modulus,

$$B_0 = V_0 \left. \frac{d^2E}{dV^2} \right|_{V_0}, \tag{3.1}$$

where V_0 is the equilibrium volume. Consequently, the curvature provides information about the stiffness of the predicted structure. Although no quantitative comparison of the fitted bulk moduli is performed here, the DFT and fine-tuned EOS curves exhibit qualitatively similar curvature near their respective minima.

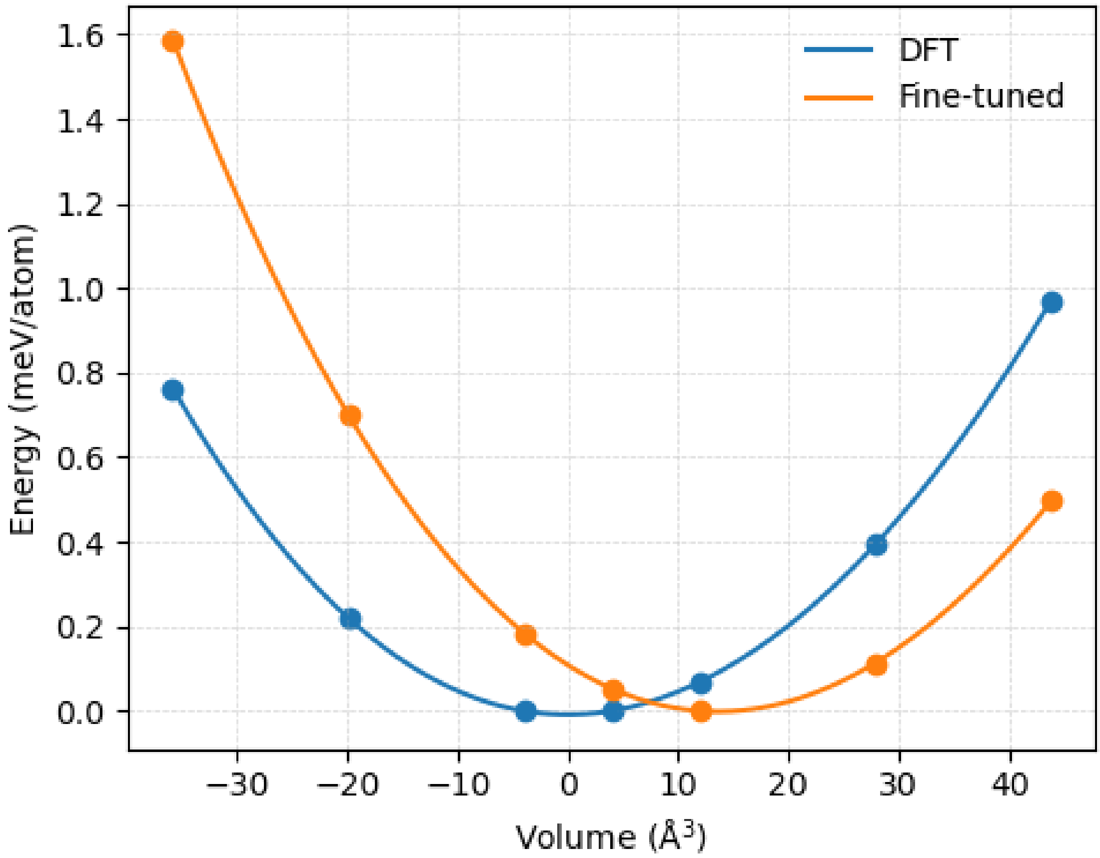

Figure 3.8: Overlay of equation of state (EOS) curves for CsPbBr from DFT (blue) and the 200-shot fine-tuned model (orange), showing energy (meV/atom) as a function of volume (Å^3) relative to the equilibrium configuration. Note the close agreement in the curvature as well as the shift in the minima.

This suggests that the fine-tuned model captures the local shape of the energy landscape around equilibrium reasonably well, even though the position of the minimum is shifted.

Additional structures from the EOS analysis were added to the dataset for fine-tuning to test whether the volume discrepancy arose from insufficient sampling of compressed and expanded configurations. The resulting volume trends remained similar, suggesting that the discrepancy is unlikely to be caused only by a lack of compositional or volumetric diversity in the training data. This does not identify the exact origin of the shift, but it narrows the explanation: the difference appears to be a systematic feature of the fine-tuned potential rather than a simple failure caused by missing configurations in the dataset.

3.2.5 Discussion of Fine-Tuned Model Behaviour

A consistent observation across the fine-tuned models is that their predicted volumes are closer to experiment than to the DFT-relaxed volumes, even though the models were fine-tuned using DFT data. This behaviour should be interpreted carefully. It should not be stated as definitive proof that the model has "corrected" DFT, because the fine-tuned model is still ultimately trained on DFT-derived energies, forces, and stresses. A plausible hypothesis is that the final model reflects a balance between the pretrained foundation-model prior and the system-specific DFT fine-tuning data.

The fine-tuning does not completely overwrite the pretrained potential energy surface. The MACE-MPA model was trained on a broad and diverse materials dataset before being adapted to the present $CsPb(Cl_{1-x}Br_x)_3$ structures. Therefore, when the fine-tuning dataset is relatively small, the final model may retain some structural preferences from the pretrained model. This could shift the equilibrium volume away from the specific DFT reference used in this work, particularly if the pretrained model has learned an average structural behaviour across many DFT calculations and chemical environments.

A second possible explanation is related to the known sensitivity of DFT equilibrium geometries to methodological choices, including the exchange–correlation functional, dispersion corrections, pseudopotentials, and convergence settings. Therefore, the DFT curve used here should be treated as a reference from a specific computational setup, not as an absolute truth. The fact that the fine-tuned model lies closer to experiment may therefore be physically meaningful, but not a proof that the MLIP is more accurate than DFT for the system of interest.

The EOS results help clarify the nature of the discrepancy. The fine-tuned model does not appear to produce random or unstable energy–volume behaviour. Instead, it gives a smooth EOS curve with a well-defined minimum and a curvature similar to DFT, but with the minimum shifted to larger volume. This suggests that the model has learned a coherent local potential energy surface, while predicting a different equilibrium volume. The discrepancy is therefore better described as a systematic shift in the equilibrium minimum rather than as a failure of structural relaxation.

Overall, the fine-tuned MACE-MPA model successfully reproduces the main equilibrium structural trends of $CsPb(Cl_{1-x}Br_x)_3$: the near-linear increase in volume with Br fraction, the smooth evolution of lattice parameters, and equilibrium volumes close to experimental values. However, further work should test whether the discrepancy persists for different DFT functionals, larger fine-tuning datasets, and finite-temperature structural properties.

3.3 Octahedral Tilting

3.3.1 Structural Importance of Octahedral Tilting

Octahedral tilting plays an important role in determining the relaxed crystal symmetry of halide perovskites and therefore provides a sensitive test of whether an interatomic potential correctly captures the low-energy structural landscape. In Cs-based halide perovskites, the ideal cubic structure is often unstable at low temperatures and relaxes into lower-symmetry phases through cooperative rotations of the PbX_6 octahedra. These distortions modify the relative orientation of neighbouring octahedra and lower the total free energy of the structure.

The resulting tilt systems are commonly described using Glazer notation [50], in which the letters a, b, and c denote octahedral rotations about the crystallographic axes, while the superscripts describe the phase relationship between neighbouring octahedra. A superscript '+' denotes in-phase rotations, '−' denotes out-of-phase rotations, and '0' corresponds to no rotation about that axis. The experimentally observed low-temperature orthorhombic phase of $CsPbCl_3$ and $CsPbBr_3$ is characterised by the tilt system

$$a^-a^-c^+$$

which corresponds to out-of-phase rotations about the a and b axes and in-phase rotations about the c axis.

From a validation perspective, reproducing the correct octahedral tilt pattern is considerably more demanding than reproducing lattice constants or equilibrium volumes alone. In the present work, structures were initialised from an untilted perovskite framework, meaning that the model must relax away from the ideal cubic symmetry toward the experimentally observed tilted minimum. A model may reproduce average structural dimensions reasonably well while still failing to capture the correct tilting pattern of the octahedra. Consequently, octahedral tilting provides a more stringent benchmark of transferability and structural accuracy than volume trends alone.

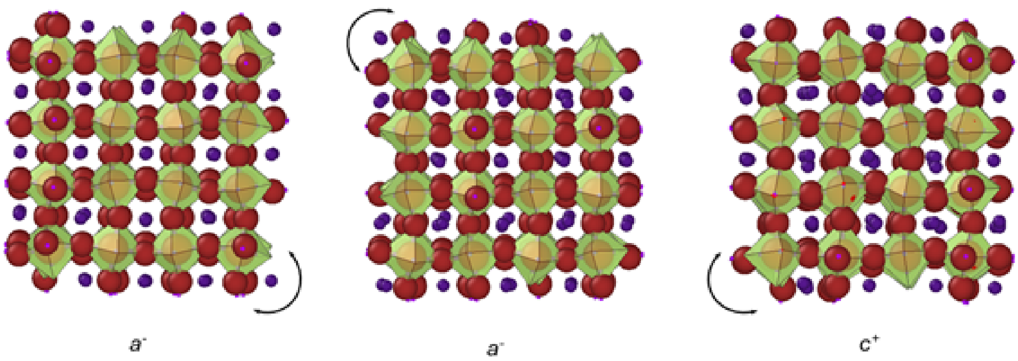

Figure 3.9: Octahedral tilting pattern in $CsPbCl_3$, illustrating the $a^-a^-c^+$ Glazer tilt system. The $PbCl_6$ octahedra (shown in green) undergo out-of-phase rotations about the a and b axes (a^-a^-) and in-phase rotations about the c axis (c^+), as indicated by the arrows. Cs atoms occupy the interstitial sites, while Cl atoms form the octahedral framework surrounding Pb.

Octahedral tilting is also structurally significant because it directly influences local bonding geometry. Cooperative rotations alter the Pb–X–Pb bond angles away from the ideal 180° configuration, modifying orbital overlap and therefore affecting the electronic structure of the material [52]. In mixed-halide systems such as $CsPb(Cl_{1-x}Br_x)_3$, the local halide environment additionally influences the magnitude and stability of the tilts due to differences in ionic radii and local strain [53]. Analysing tilt behaviour therefore provides insight into how compositional disorder affects the structural stability of the perovskite framework.

Time Evolution of Octahedral Tilt Angles

To further analyse the stability of the tilted perovskite phase, the time evolution of the octahedral tilt angles was monitored throughout the molecular dynamics simulations using the fine-tuned model. The instantaneous tilt angles along the crystallographic a, b, and c directions are shown in Fig. 3.10. Shortly after the beginning of the trajectory, the system develops finite octahedral rotations, indicating a rapid relaxation away from the initially untilted cubic configuration and into a lower-symmetry tilted structure.

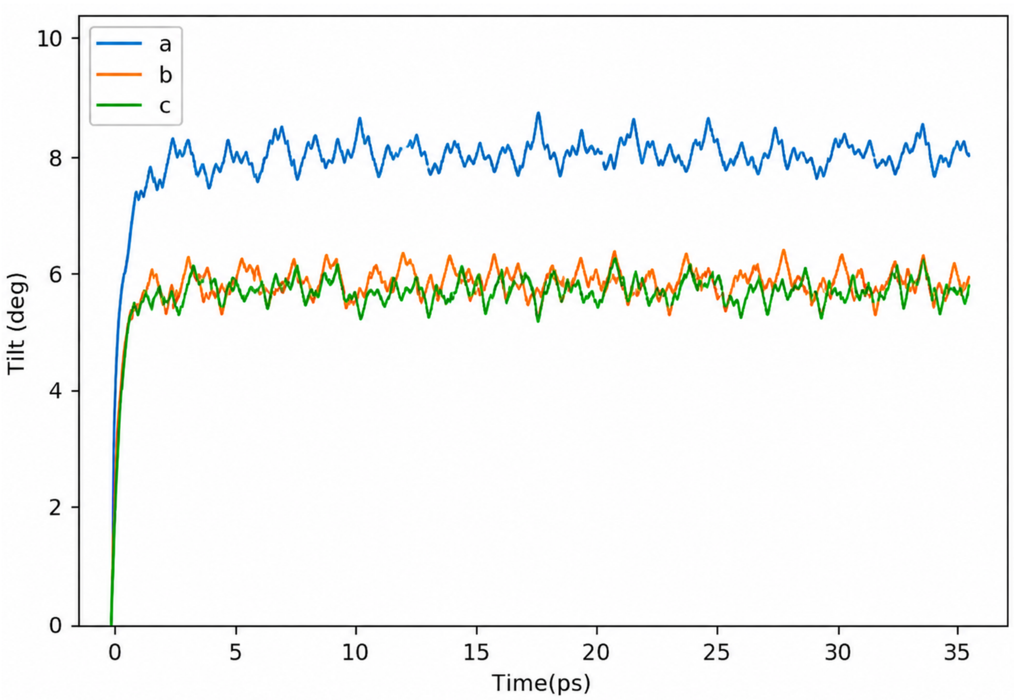

Figure 3.10: Time evolution of octahedral tilt angles (in degrees) along the a, b, and c directions during molecular dynamics simulations. The panel shows the instantaneous tilt angles as a function of time, highlighting thermal fluctuations around equilibrium values.

Following this initial transient regime, the tilt angles fluctuate around stable non-zero average values of approximately 6°–8°. These fluctuations are expected at finite temperature and arise from thermal motion of the octahedra about their equilibrium orientations. Importantly, the tilt magnitudes do not decay back toward zero over the duration of the simulation, indicating that the tilted phase remains dynamically stable throughout the trajectory.

The time-averaged tilt angles, shown as dashed lines in Fig. 3.10, further confirm the persistence of the distorted structure. Although the instantaneous values exhibit thermal fluctuations, the average tilt amplitudes remain relatively constant over time, demonstrating that the system fluctuates around a well-defined tilted minimum rather than sampling an untilted cubic phase. The anisotropy between the different crystallographic directions is also consistent with the experimentally observed orthorhombic tilt system.

These results provide evidence that the fine-tuned model not only reproduces the correct static octahedral tilt pattern after structural relaxation, but is also capable of maintaining the tilted phase dynamically during finite-temperature simulations. The persistence of stable non-zero tilt angles reflects the underlying stability of the low-symmetry energy minimum captured by the potential energy surface.

Tilt Correlation Polarity

While the tilt angles describe the magnitude of the octahedral rotations, the tilting correlation polarity (TCP) provides information about the phase relationship between neighbouring octahedra and therefore allows direct identification of the Glazer tilt system. Following the definition implemented in PDynA [54],

$$\delta_\alpha = \frac{n_\alpha^+ - n_\alpha^-}{n_\alpha^+ + n_\alpha^-}, \tag{3.2}$$

where n_α^+ and n_α^- are the numbers of positively and negatively correlated neighbouring octahedral pairs along the crystallographic direction α, respectively. The correlation parameter ranges from -1 to $+1$, with positive values corresponding to in-phase tilting, negative values corresponding to out-of-phase tilting, and values near zero indicating the absence of long-range tilt correlation. Consequently, δ_α provides a quantitative measure of the degree and polarity of octahedral tilt ordering along a given crystallographic direction.

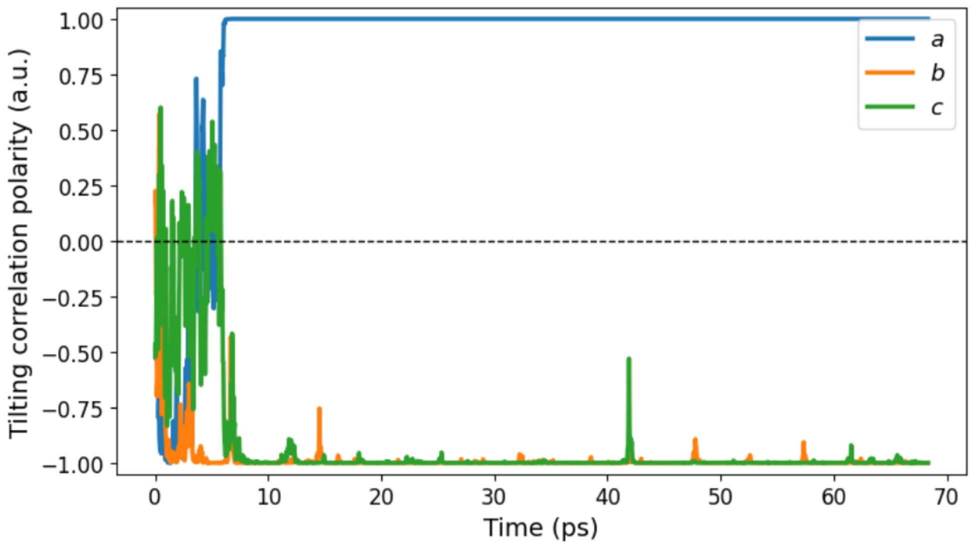

Figure 3.11: Time evolution of octahedral tilting correlation polarity along the a, b, and c directions. The correlation parameter (dimensionless) indicates whether neighboring octahedra tilt in-phase (positive values) or out-of-phase (negative values), with zero representing no correlation.

Table 3.2 shows the time evolution of the tilt-correlation polarity along the three crystallographic directions. During the initial stages of the trajectory, the correlation values fluctuate significantly as the system relaxes away from the initially symmetric configuration. After this transient regime, the correlations stabilise into well-defined positive and negative values corresponding to persistent rotational ordering of the octahedra.

The results show that one crystallographic direction is in-phase, while the remaining ones are out-of-phase. This behaviour is consistent with the experimentally observed $a^-a^-c^+$ Glazer tilt system for orthorhombic $CsPbX_3$ perovskites.

The persistence of these correlation polarities over the duration of the simulation demonstrates that the rotational topology of the octahedral framework is maintained dynamically and is not simply a transient feature of the structural relaxation.

Overall, the combined analysis of tilt-angle evolution and tilt-correlation polarity demonstrates that the fine-tuned potential successfully reproduces both the magnitude and the topology of the experimentally observed octahedral tilting behaviour. This provides strong evidence that the model captures the collective structural physics governing the low-symmetry phases of mixed metal-halide perovskites.

3.3.2 Comparison of Experimental and Computational Tilt Predictions

Table 3.2 compares the experimentally observed octahedral tilt systems of $CsPbBr_3$ and $CsPbCl_3$ with those predicted by the different foundational and fine-tuned machine learning interatomic potentials.

(a) $CsPbBr_3$

	Lattice parameters												
	a (Å)	Δ (%)	b (Å)	Δ (%)	c (Å)	Δ (%)	α (°)	Δ (%)	β (°)	Δ (%)	γ (°)	Δ (%)	Tilts
Experimental	8.203 (18)	-	11.753 (22)	-	8.250 (21)	-	90.0	-	90.0	-	90.0	-	$a^-a^-c^+$
mpa	8.273	0.85	11.896	1.21	8.481	2.80	90.0	0.0	90.0	0.0	90.0	0.0	$a^-a^-c^-$
omat-0	8.23	0.32	11.908	1.32	8.55	3.63	90.0	0.0	90.0	0.0	90.0	0.0	$a^-a^-c^-$
mp-0a	8.276	0.88	11.97	1.84	8.536	3.46	90.0	0.0	90.0	0.0	90.0	0.0	$a^-b^-c^-$
ob3	7.937	3.25	11.47	2.41	8.065	2.24	90.0	0.0	90.0	0.0	90.0	0.0	$a^-b^-c^+$
fine-tuned	8.241	0.45	11.70	-0.81	8.282	0.38	90.0	0.0	90.0	0.0	90.0	0.0	$a^-a^-c^+$

(b) $CsPbCl_3$

	Lattice parameters												
	a (Å)	Δ (%)	b (Å)	Δ (%)	c (Å)	Δ (%)	α (°)	Δ (%)	β (°)	Δ (%)	γ (°)	Δ (%)	Tilts
Experimental	7.899 (14)	-	11.247 (11)	-	7.901 (14)	-	90.0	-	90.0	-	90.0	-	$a^-a^-c^+$
mpa	7.911	0.15	11.342	0.84	8.191	2.54	90.0	0.0	90.0	0.0	90.0	0.0	$a^-a^-c^+$
omat-0	7.951	0.65	11.395	1.31	8.132	2.91	90.0	0.0	90.0	0.0	90.1	0.1	$a^-b^-c^-$
mp-0a	7.956	1.22	11.388	1.32	8.142	2.82	90.0	0.0	90.0	0.0	90.0	0.0	$a^0a^0a^0$
ob3	7.937	0.98	11.398	1.41	8.132	2.69	90.0	0.0	90.0	0.0	90.0	0.0	$a^0b^0c^0$
fine-tuned	7.913	0.17	11.301	0.48	7.963	0.78	90.0	0.0	90.0	0.0	90.0	0.0	$a^-a^-c^+$

Table 3.2: Comparison of lattice parameters (a, b, and c in Å), cell angles (α, β, and γ in degrees), and octahedral tilt systems for orthorhombic $CsPbCl_3$ and $CsPbBr_3$ obtained from experiment and different computational models, including foundational and fine-tuned machine-learning interatomic potentials. Percentage deviations (Δ%) from experimental values are reported for each structural parameter. The highlighted rows indicate the best overall agreement with experiment, with the 200-shot fine-tuned model achieving the closest simultaneous match to both the equilibrium lattice parameters and the low-symmetry rotational ordering of the PbX_6 octahedral framework.

For $CsPbBr_3$, only the fine-tuned model reproduces the experimentally observed

$a^-a^-c^+$ tilt pattern. In contrast, MPA and OMAT predict an $a^-a^-c^-$ tilt pattern, while MP-0a predicts an $a^-b^-c^-$ configuration. Although MPA reproduces the lattice constants with reasonable accuracy, it does not necessarily recover the correct low-symmetry rotational ordering of the $PbBr_6$ octahedra. This demonstrates that accurate prediction of equilibrium lattice dimensions alone is insufficient to guarantee a correct description of the octahedral framework.

Differences also exists in $CsPbCl_3$. While the MPA and fine-tuned models again correctly reproduce the experimental $a^-a^-c^+$ tilt system, the remaining foundational models predict significantly different rotational orderings. The OMAT model predicts an $a^-b^-c^-$-type distortion, implying out-of-phase tilting along all three crystallographic directions. The MP-0a model instead predicts an approximately untilted or weakly distorted structure ($a^0a^0a^0$ configuration), while the OB3 model similarly fails to recover the experimentally observed cooperative tilt ordering. These discrepancies indicate that several of the foundational models are unable to correctly reproduce the subtle symmetry-breaking distortions present in the chloride system.

The fine-tuned model shows the most consistent overall behaviour across both compounds. In addition to producing lattice parameters closer to experiment, it also correctly reproduces the experimentally observed $a^-a^-c^+$ tilt system in both $CsPbBr_3$ and $CsPbCl_3$. This demonstrates that the fine-tuning procedure improves not only average structural quantities such as equilibrium volume, but also the collective rotational ordering of the octahedral network. Since octahedral tilting emerges from subtle cooperative interactions between neighbouring octahedra, correctly reproducing the tilt topology indicates that the model captures the symmetry-lowering physics governing the low-energy structural landscape.

Importantly, the comparison also highlights why tilt prediction is a more sensitive structural benchmark than volume trends alone. Several of the foundational models produce lattice constants with relatively small percentage deviations from experiment while still predicting incorrect tilt patterns.

3.4 Finite-Temperature Molecular Dynamics and Phase Stability

3.4.1 Finite-Temperature Molecular Dynamics of $CsPbCl_3$

Understanding the finite-temperature behaviour of inorganic halide perovskites is essential because their structural phases are strongly coupled to thermal fluctuations, octahedral tilting, and anharmonic lattice dynamics. This also relates to the overall aim of the work to have a general potential for the mixed compositions in order to understand phase transitions with varying mixed halide compositions. Static temperature relaxations alone are insufficient to capture the experimentally observed phase evolution of $CsPbCl_3$, which undergoes a sequence of orthorhombic, tetragonal, and cubic structural transitions with increasing temperature. Previous studies based on machine-learned interatomic potentials have shown that these transitions are highly sensitive to simulation methodology, including heating rates, system size, and the quality of the underlying potential energy surface [12, 55, 56].

In this work, molecular dynamics simulations were performed on the $CsPbCl_3$ end member in order to evaluate whether foundational and fine-tuned machine learning models can reproduce the expected temperature-dependent structural evolution. The end-member system was selected because it provides a controlled benchmark for assessing the physical reliability of the models prior to extending the analysis to mixed-halide compositions, where compositional disorder introduces additional complexity.

Heating and cooling-ramp simulations were carried out from 1 K to 400 K and back to 1 K for 200 ps each using both the foundational and fine-tuned models.The initial heating and cooling simulations were performed using a relatively small $2 \times 2 \times 2$ supercell in order to enable rapid exploratory analysis of the finite-temperature phase behaviour while maintaining manageable computational cost.

Using a smaller system allowed efficient testing of the foundational and fine-tuned models and preliminary identification of temperature-dependent structural trends prior to the constant temperature runs performed later. Reduced lattice parameters were extracted throughout the trajectories in order to monitor structural evolution and identify phase transitions.

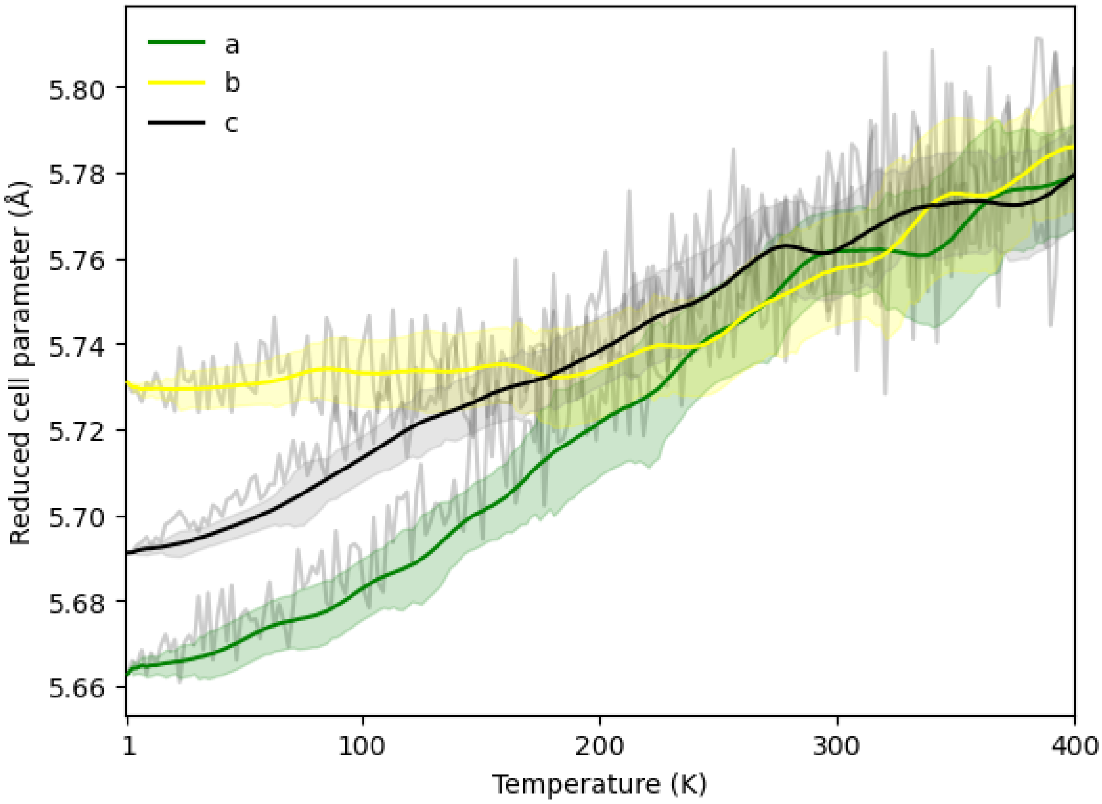

Figure 3.12: Temperature dependence of the reduced lattice parameters (a, b, and c) obtained during a heating simulation of $CsPbCl_3$ using the MPA interatomic potential. The coloured curves represent smoothed trends for the three crystallographic directions, while the grey trajectories show the instantaneous fluctuations from the MD simulation. The variance bands are also shown.

From Fig 3.12, the foundational model reproduced the expected qualitative phase sequence, beginning in the orthorhombic phase at low temperature, transitioning toward a tetragonal phase near intermediate temperatures, and approaching cubic-like behaviour at elevated temperatures. This behaviour is consistent with previous experimental and computational studies of $CsPbCl_3$, which report progressive suppression of octahedral tilting with increasing temperature.[9]

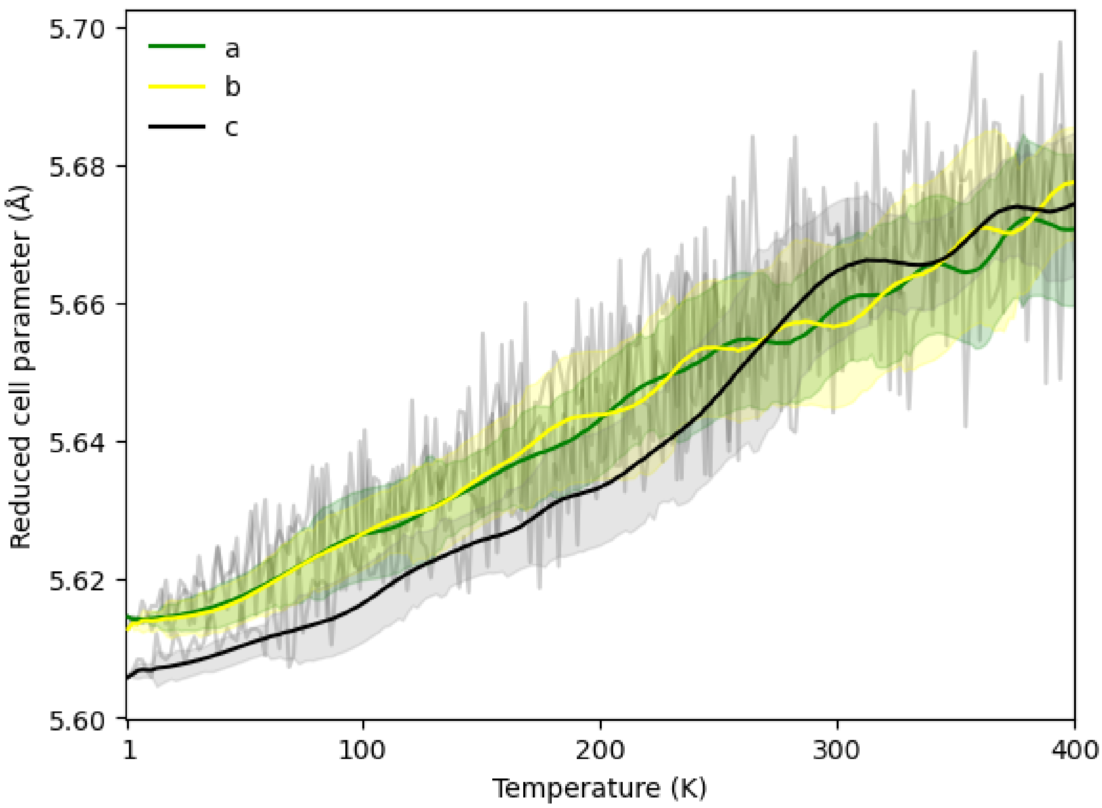

Figure 3.13: Temperature dependence of the reduced lattice parameters (a, b, and c) obtained during a heating molecular dynamics simulation of $CsPbCl_3$ using the fine-tuned MACE interatomic potential. The coloured curves represent smoothed trends for the three crystallographic directions, while the grey trajectories show the instantaneous fluctuations from the MD simulation.The variance bands are also shown.

In contrast, from Fig 3.13, the fine-tuned model displayed significantly different dynamical behaviour. Although the model reproduced structural parameters with improved static accuracy, the low-temperature orthorhombic phase was not stabilised. Instead, the trajectory remained predominantly tetragonal at low temperature before transitioning toward a cubic-like phase at higher temperature (400 K). This indicates that fine-tuning improved agreement with static structural properties while changing the temperature-dependent phase behaviour predicted by the model.

This distinction highlights an important limitation of purely structural validation. Accurate lattice parameters alone do not guarantee that the model correctly reproduces the anharmonic potential energy surface responsible for temperature-driven phase transitions. The results therefore suggest that preserving dynamical behaviour is a more difficult criterion than reproducing equilibrium structural observables in isolation.

Furthermore, the observed discrepancies with the fine-tuned model are consistent with previous literature demonstrating that finite-temperature phase transitions in halide perovskites are highly sensitive to the details of the potential energy surface and simulation protocol. Fransson *et al.* showed that MLIPs trained using different exchange-correlation functionals can predict significantly different transition temperatures and phase behaviour in $CsPbX_3$ systems. Additionally, the results depend on factors such as heating/cooling rate and simulation size [12].

3.4.2 Structural Validation, Statistical Convergence, and Dynamical Reliability

To quantitatively evaluate the finite-temperature structural performance of the different MLIPs, the molecular dynamics trajectories were analysed using three complementary descriptors: phase stability, octahedral tilting behaviour, and lattice-parameter accuracy. These descriptors were combined into an overall structural reliability score in order to distinguish models that reproduce only average structural properties from those that correctly capture the underlying finite-temperature lattice dynamics.

For each trajectory, the instantaneous simulation cell lengths were extracted and converted into reduced pseudocubic lattice parameters according to

$$a_{\mathrm{red}} = \frac{a_{\mathrm{cell}}}{8\sqrt{2}}, \qquad b_{\mathrm{red}} = \frac{b_{\mathrm{cell}}}{8\sqrt{2}}, \qquad c_{\mathrm{red}} = \frac{c_{\mathrm{cell}}}{8}.$$

This transformation enables direct comparison between distorted orthorhombic, tetragonal, and cubic structures within a common pseudocubic framework.

The instantaneous symmetry breaking of the lattice was quantified using the pairwise lattice-parameter differences

$$\Delta_{ab} = |a_{\text{red}} - b_{\text{red}}|, \qquad \Delta_{bc} = |b_{\text{red}} - c_{\text{red}}|, \qquad \Delta_{ac} = |a_{\text{red}} - c_{\text{red}}|.$$

The tolerance was computed from the distribution of all pairwise lattice differences using the 50th percentile,

$$\delta_{\text{tol}} = P_{50}\left(\Delta_{ab}, \Delta_{bc}, \Delta_{ac}\right),$$

where P_{50} denotes the median of the combined lattice-difference distribution. The use of the median tolerance provides a more conservative and physically stable classification criterion than higher-percentile thresholds, which tended to over-classify thermally fluctuating configurations as cubic.

Using this tolerance, each snapshot was classified into one of four structural categories:

- cubic: $a \approx b \approx c$,
- tetragonal: $a \approx b$, but c distinct,
- orthorhombic: all three lattice parameters distinct,
- distorted: intermediate configurations that do not satisfy the symmetry conditions above.

The distorted class was introduced as thermally fluctuating perovskite structures frequently occupy transient intermediate states that cannot be rigorously assigned to ideal crystallographic symmetries. Rather than forcing these configurations into discrete phase labels, the distorted category prevents artificial over-classification and provides a more physically realistic representation of anharmonic finite-temperature dynamics.

For each temperature, the phase agreement was quantified as

$$p_T = \frac{N_{\text{correct}}(T)}{N_{\text{total}}(T)},$$

where $N_{\text{correct}}(T)$ is the number of snapshots assigned to the experimentally expected phase at temperature T.

To penalise unstable phase behaviour, the phase score additionally incorporated the statistical variance of the phase assignment,

$$\sigma_T = \sqrt{p_T(1 - p_T)},$$

The averaged phase score across all temperatures was then computed as

$$\overline{S}_{\text{phase}} = \frac{1}{3}\left[S_{\text{phase}}(100\,\text{K}) + S_{\text{phase}}(250\,\text{K}) + S_{\text{phase}}(400\,\text{K})\right].$$

The lattice-parameter agreement was evaluated using

$$S_{\text{lattice}}(T) = \exp[-K_{\text{lattice}}(T)],$$

where

$$K_{\text{lattice}}(T) = \frac{1}{6}\sum_i \frac{|i^{\text{model}}(T) - i^{\text{exp}}(T)|}{i^{\text{exp}}(T)}.$$

where the summation was performed over the six structural parameters $i = \{a, b, c, \alpha, \beta, \gamma\}$,

Because reliable experimental tetragonal lattice parameters at 250 K were unavailable from the ICSD datasets used in this work, the lattice score was averaged only over 100 K and 400 K,

$$\overline{S}_{\text{lattice}} = \frac{1}{2}\left[S_{\text{lattice}}(100\,\text{K}) + S_{\text{lattice}}(400\,\text{K})\right].$$

Agreement with the experimentally expected octahedral tilting pattern was quantified using

$$S_{\text{tilt}} = 1 - \frac{1}{3}\sum_{j=1}^{3} d_j,$$

where d_j represents the mismatch in tilt behaviour along each crystallographic direction.

Finally, the overall structural score was defined as

$$S_{\text{score}} = \frac{1}{3}\overline{S}_{\text{phase}} + \frac{1}{3}S_{\text{tilt}} + \frac{1}{3}\overline{S}_{\text{lattice}}.$$

This combined metric separates three distinct aspects of structural reliability: phase stability, octahedral tilt symmetry, and quantitative lattice accuracy. Such separation is important because a model may reproduce average lattice parameters accurately while still failing to stabilise the correct finite-temperature phase behaviour.

Because molecular dynamics trajectories are temporally correlated, neighbouring snapshots cannot be treated as statistically independent samples. Reliable extraction of equilibrium structural properties therefore requires explicit analysis of convergence behaviour, autocorrelation times, and statistical uncertainty.

To assess equilibration, both running averages and rolling-window averages of the lattice score were analysed throughout the trajectories. Early sections of the simulations exhibited transient fluctuations associated with thermal equilibration and initial structural relaxation. Only the equilibrated region of the trajectory was retained for final statistical analysis.

The autocorrelation function of the lattice score was then computed as

$$C(k) = \frac{\langle (S_t - \bar{S})(S_{t+k} - \bar{S}) \rangle}{\sigma^2},$$

where k is the lag time, $\bar{S}$ is the equilibrium average lattice score, and σ is the standard deviation of the lattice score.

$$\tau_{\text{int}} = 1 + 2\sum_{k>0} C(k),$$

provides a measure of the timescale over which the trajectory remains correlated. Large autocorrelation times indicate slower structural relaxation and reduced numbers of statistically independent configurations.

The integrated autocorrelation time was additionally used to determine the moving-block bootstrap size. Rather than resampling individual snapshots independently, contiguous blocks of trajectory data with length

$$L_{\text{block}} = 2\tau_{\text{int}}$$

were resampled in order to preserve the intrinsic temporal correlations of the molecular dynamics trajectories.

For each constant-temperature trajectory, 5000 bootstrap resamplings were performed to estimate the statistical uncertainty in the equilibrium lattice score. The resulting uncertainties were found to be extremely small, typically on the order of 10^{-5}, indicating strong statistical convergence of the ensemble averages.

The combination of convergence analysis, autocorrelation analysis, and moving-block bootstrap uncertainty estimation therefore provides a statistically rigorous framework for interpreting the finite-temperature structural behaviour of the models. Importantly, the observed differences between models originate primarily from genuine differences in the learned potential energy surfaces rather than stochastic sampling noise.

The results reveal several important trends regarding the relationship between fine-tuning, structural accuracy, and finite-temperature dynamics. First, the lattice scores remain extremely high and nearly identical across all models, with only very small variations between 0.986 and 0.988. This indicates that all models reproduce

Table 3.3: Comparison of the structural reliability metrics for the foundational MPA model and the fine-tuned models.

Model	$\mathbf{S}_{\text{phase}}$	$\mathbf{S}_{\text{tilt}}$	$\mathbf{S}_{\text{lattice}}$	$\mathbf{S}_{\text{score}}$	$\boldsymbol{\tau}_{\text{int}}$ **(100 K)**	$\boldsymbol{\tau}_{\text{int}}$ **(400 K)**
mpa	0.1855 ± 0.005	2/3	0.98667 ± 0.00002	0.6074 ± 0.0006	7.49	12.06
200-shot	0.2854 ± 0.004	1	0.98766 ± 0.00002	0.7611 ± 0.0004	12.98	46.02
100-shot	0.1971 ± 0.003	1	0.98790 ± 0.00002	0.7316 ± 0.0003	15.66	49.17
50-shot	0.2591 ± 0.003	2/3	0.98786 ± 0.00002	0.6410 ± 0.0003	15.03	51.99
20-shot	0.2163 ± 0.004	1/3	0.98786 ± 0.00002	0.5157 ± 0.0004	15.18	53.46

the average lattice dimensions with similarly high accuracy. The low bootstrap uncertainties further confirm that these averages are statistically well converged.

However, substantially larger variations are observed in the phase and tilt metrics. In particular, the 200-shot and 100-shot models achieve perfect tilt agreement, whereas the foundational MPA model reproduces only partial agreement with the expected tilt pattern.

Interestingly, the best overall structural performance is obtained for the 200-shot model, which simultaneously achieves the highest phase score, perfect tilt agreement, and the largest overall structural score. This suggests that moderate fine-tuning improves both local structural accuracy and collective octahedral behaviour without excessively distorting the underlying anharmonic energy landscape.

A non-monotonic trend is observed as the fine-tuning dataset becomes progressively smaller. While the 50-shot and 20-shot models retain excellent lattice accuracy, their phase and tilt scores deteriorate noticeably. The 20-shot model in particular exhibits the weakest tilt agreement and the lowest overall structural score. This behaviour suggests that fine-tuning on smaller datasets is not enough to capture the overall potential energy landscape.

The autocorrelation analysis provides additional physical insight into the finite-temperature dynamics. At 100 K, the integrated autocorrelation times remain relatively modest for all models, indicating comparatively rapid decorrelation and stable sampling within the orthorhombic phase. In contrast, all models exhibit dramatically larger autocorrelation times at 400 K, with values increasing by factors of approximately three to five.

This behaviour is physically consistent with the increased anharmonicity and dynamical disorder expected near the high-temperature cubic phase. At elevated temperatures, octahedral rotations and lattice fluctuations persist for longer timescales, leading to slower structural relaxation and stronger temporal correlations.

The foundational MPA model exhibits the smallest autocorrelation times at both temperatures, indicating comparatively faster structural relaxation and weaker persistence of thermal distortions. In contrast, the fine-tuned models display substantially slower dynamics at 400 K. One possible interpretation is that fine-tuning deepens local minima within the learned potential energy surface, thereby increasing the lifetime of distorted configurations and slowing transitions between competing structural motifs.

The apparent difference between the heating–cooling analysis and the fixed-temperature equilibrium metrics arises because the two protocols test different aspects of model behaviour. The heating–cooling run probes whether a model can reproduce the qualitative temperature-driven phase sequence, whereas the equilibrium analysis evaluates the stability of the expected phase at each fixed temperature after equilibration.

In the smaller-system heating–cooling simulations, the foundational MPA model reproduced the expected orthorhombic-to-tetragonal-to-cubic sequence more clearly. This suggests that the underlying potential energy surface of the foundational model preserves the broad ordering of the competing structural phases. However, in the fixed-temperature analysis, the phase score is computed from the fraction of equilibrated snapshots assigned to the expected phase.

The fine-tuned models, particularly the 200-shot model, appear to stabilise the expected fixed-temperature structures more strongly under the equilibrium protocol, leading to higher phase and overall scores. This does not necessarily imply that they reproduce the global transition pathway better than the foundational model. Rather, it indicates that under independently equilibrated simulations at selected temperatures, their local phase stability is stronger according to the present classification metric.

This distinction highlights the importance of using both protocols. Heating–cooling simulations assess qualitative transition behaviour and hysteresis, while fixed-temperature simulations provide a stricter measure of equilibrium phase stability, lattice accuracy, and statistical convergence. The difference between the two results therefore suggests that the foundational model may better preserve global phases, whereas selected fine-tuned models may better stabilise local equilibrium structures at the specific temperatures tested.

Although the present analysis cannot uniquely determine the microscopic origin of these effects, one plausible hypothesis is that smaller fine-tuning datasets reduce the diversity of collective octahedral environments sampled during training. As a result, the resulting models may reproduce local equilibrium structures accurately while failing to generalise the cooperative long-range lattice dynamics responsible for phase stability at finite temperature.

Conclusion and Future Work

In this work, foundational machine-learned interatomic potentials were benchmarked and fine-tuned in order to investigate the structural behaviour of mixed-halide $CsPb(Cl_{1-x}Br_x)_3$ perovskites. The results demonstrate that pretrained MACE-based models provide a strong starting point for modelling inorganic halide perovskites and that fine-tuning with relatively small DFT datasets substantially improves agreement with experimental structural trends.

However, several limitations and directions for future work remain. Although relatively small CUR-selected datasets were sufficient for reproducing equilibrium structural properties, larger and more diverse training datasets may improve the description of local structural distortions, finite-temperature dynamics, and phase stability. This can also help in addressing the discrepancy observed in the deviation of the results of the fine-tuned model from the DFT data (Section 3.2.5). Future studies could therefore investigate whether increasing the number of DFT-labelled structures improves the transferability and predictive accuracy of the fine-tuned models.

The present work also employed DFT reference data generated using a single exchange–correlation functional. Since equilibrium structural properties and energy landscapes in halide perovskites are sensitive to the choice of functional, future work could explore training and fine-tuning using DFT datasets generated with alternative exchange–correlation approximations.

Additionally, the simulations performed here were primarily based on relatively small supercells. Larger supercell calculations may better capture long-range octahedral correlations, configurational disorder, and collective finite-temperature fluctuations. Extending the simulations to larger systems would therefore provide a more rigorous assessment of phase-transition behaviour and dynamical stability.

Finally, future work could systematically investigate how phase-transition temperatures evolve as a function of halide composition across the $CsPb(Cl_{1-x}Br_x)_3$ series. The present study validates the finite-temperature behaviour only for the end-member compounds. Such an investigation would enable direct comparison with first-principles phase-diagram studies, which predict composition-dependent phase boundaries and significant variations in the transition temperature across the $CsPb(Br_{1-x}Cl_x)_3$ series [20]. Establishing the corresponding transition-temperature versus composition relationship from molecular dynamics simulations would provide a stringent test of the transferability of the fine-tuned potential across the full compositional range.

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